



Jetson Orin Flashing Guide Collection

Developer Kit • C1901 • C1902 • C2401 • JetPack 7

LinkZee Labs

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2. Official Developer Kit Flashing

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5. C2401 Flashing

6. JetPack 7 ISO Installation

Note: Flashing parameters vary by carrier board, module, and JetPack release. Follow the matching chapter.

Install Ubuntu VM and SDK Manager

NVIDIA's official **SDK Manager** is the most common tool for flashing Jetson modules, and it needs to run on **Ubuntu**. This guide explains how to install Ubuntu inside a **VMware virtual machine** and deploy SDK Manager.

Recommended Setup

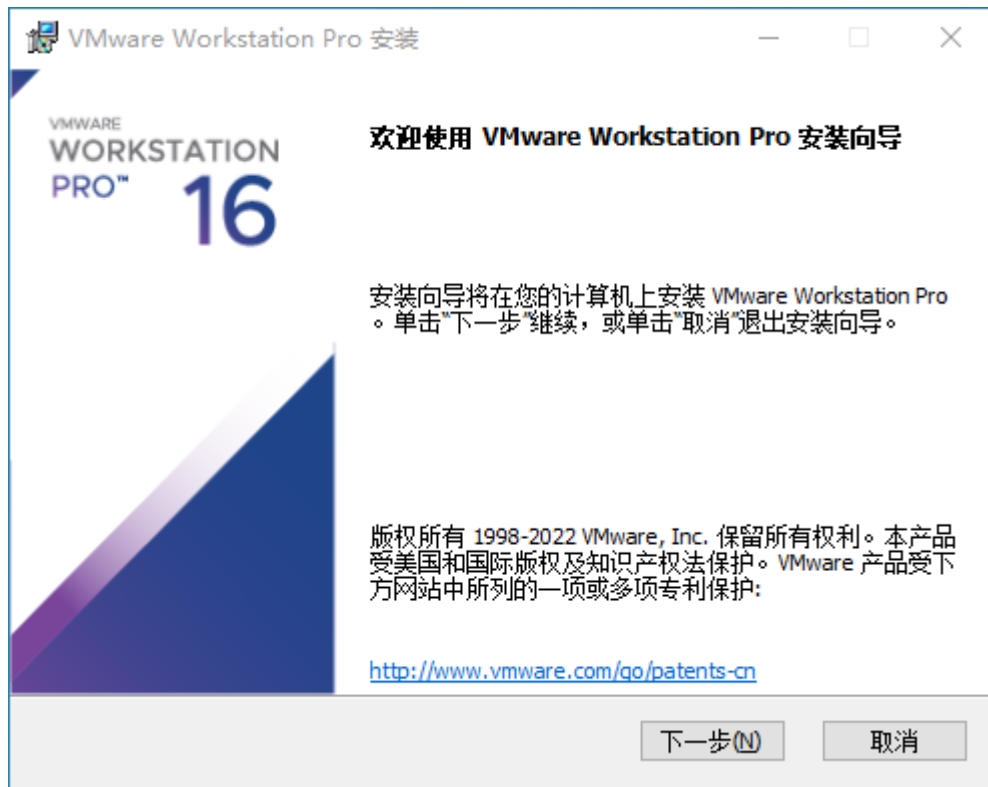
- Ubuntu 20.04 or 22.04 (64-bit)
- VM memory $\geq$ 8GB, disk $\geq$ 60GB
- Reserve enough USB controllers to detect the module

1. Preparation

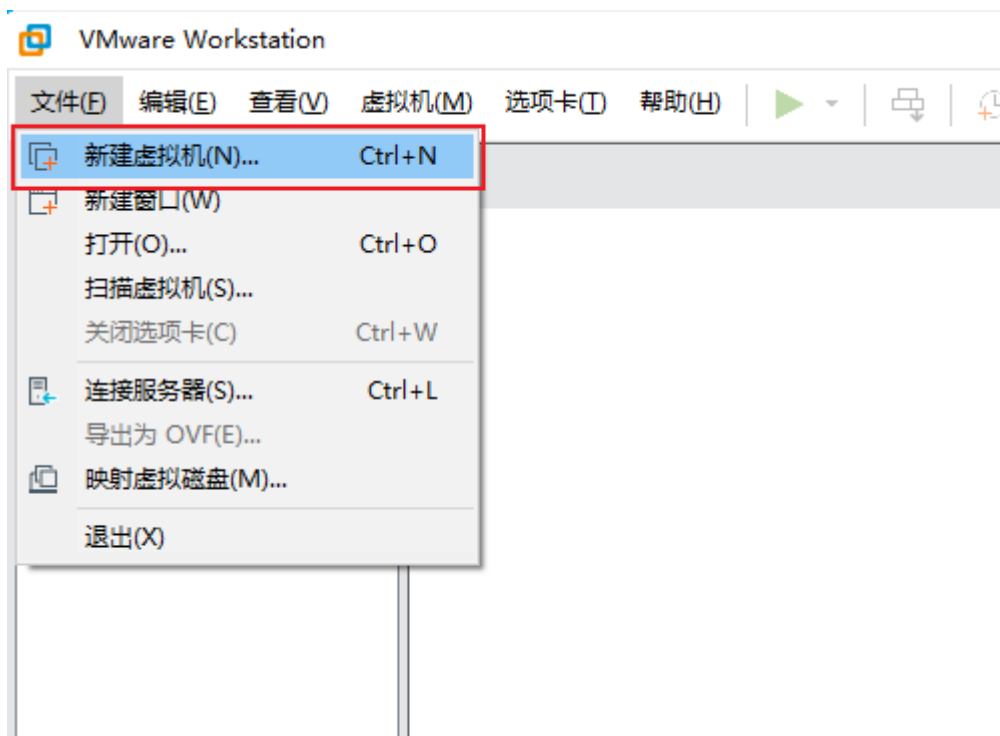
Download the following software:

Software	Description
VMware Workstation	Virtual machine software (free for personal use)
Ubuntu 20.04/22.04 ISO	System image
NVIDIA SDK Manager	Downloaded from the NVIDIA website (account required)

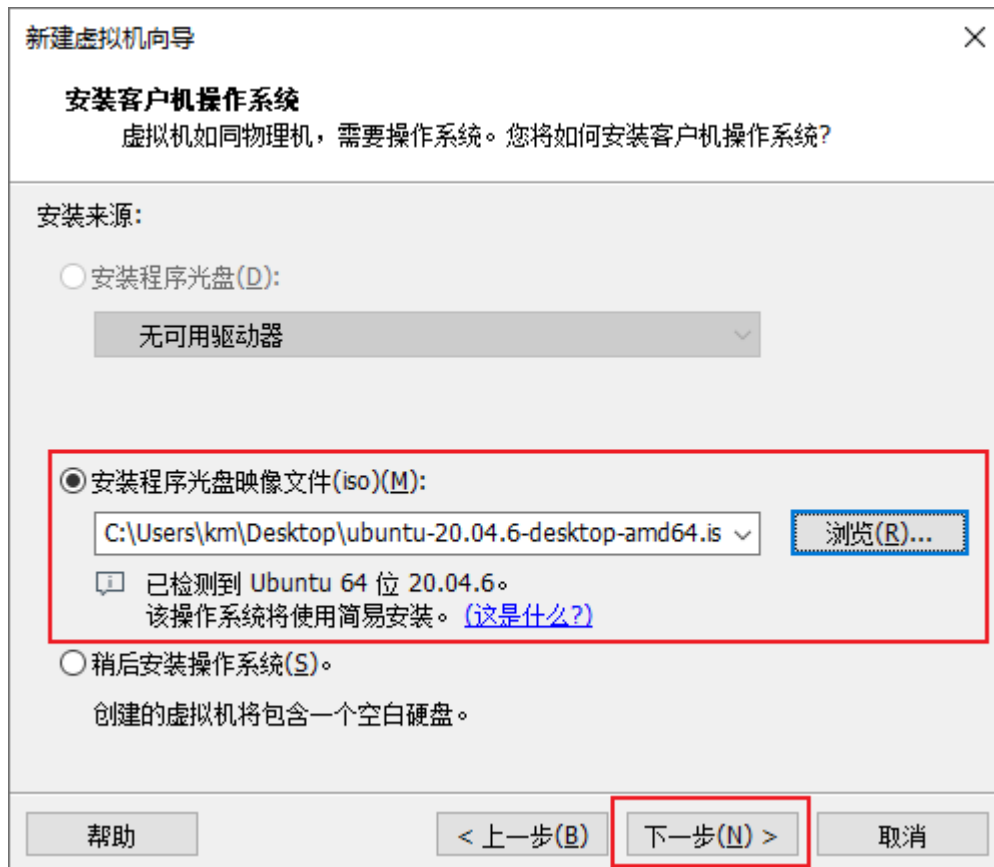
2. Install the VMware Virtual Machine



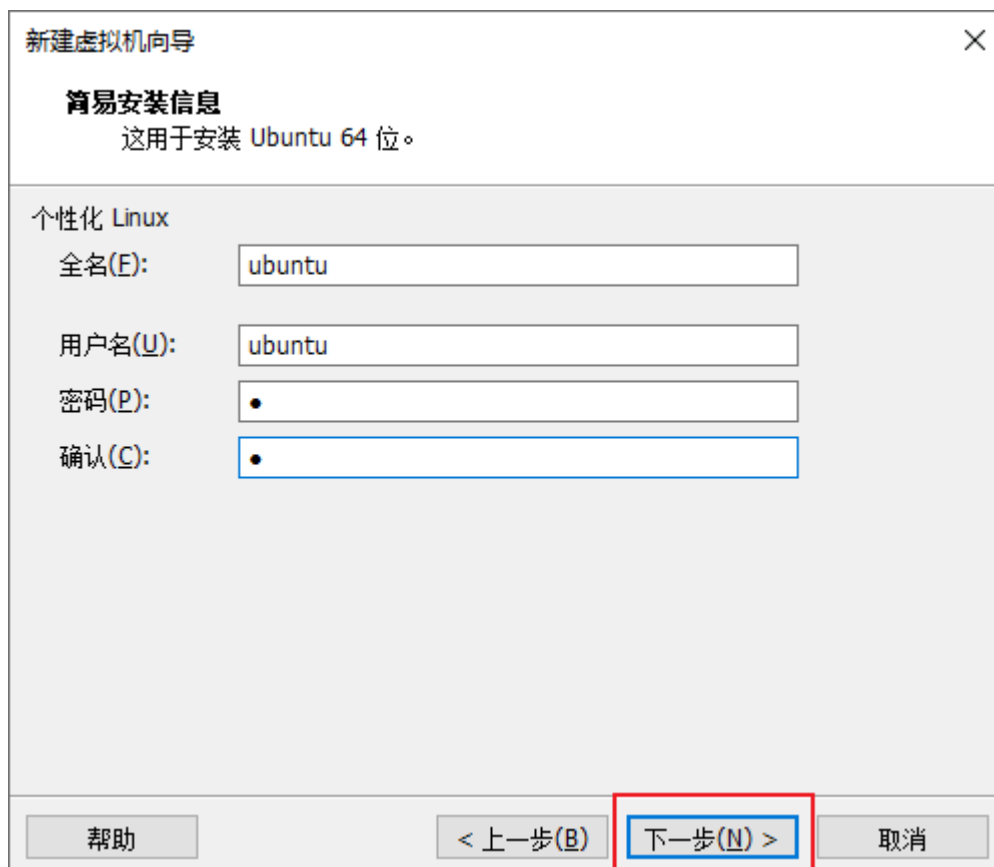
Follow the installation wizard to complete the VMware installation, then create a new virtual machine.



Select the Ubuntu ISO image you downloaded.



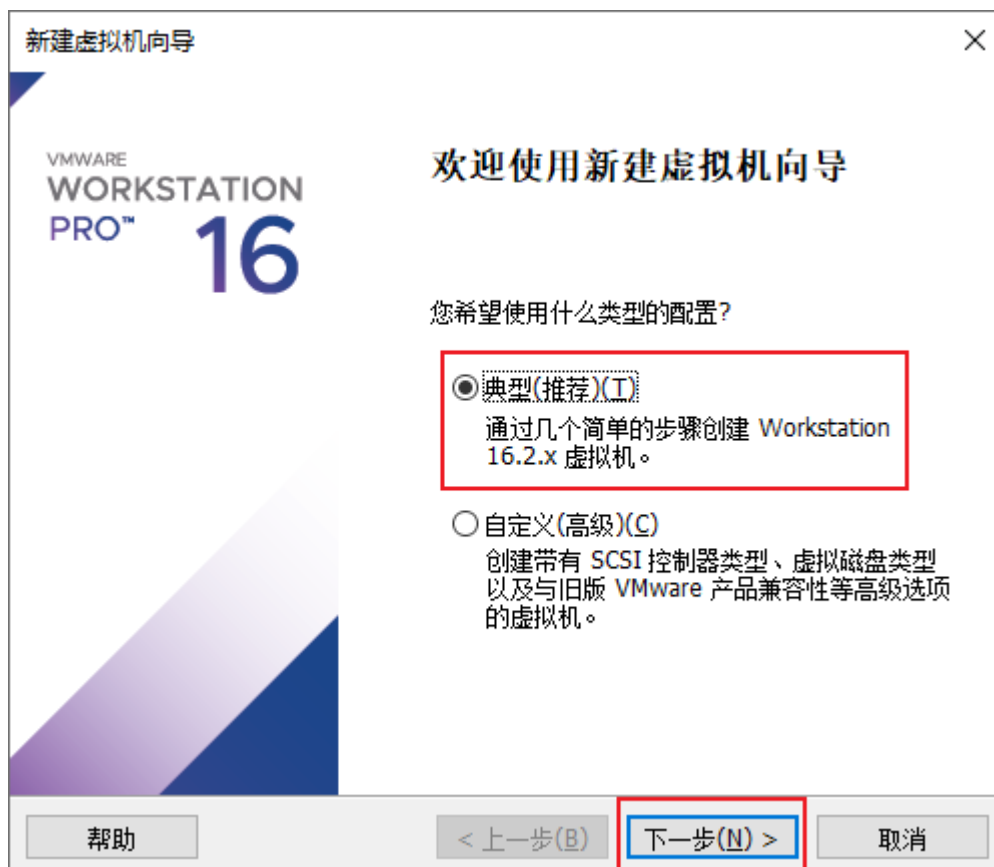
Set a username and password. Remember this password — you will use it frequently later.



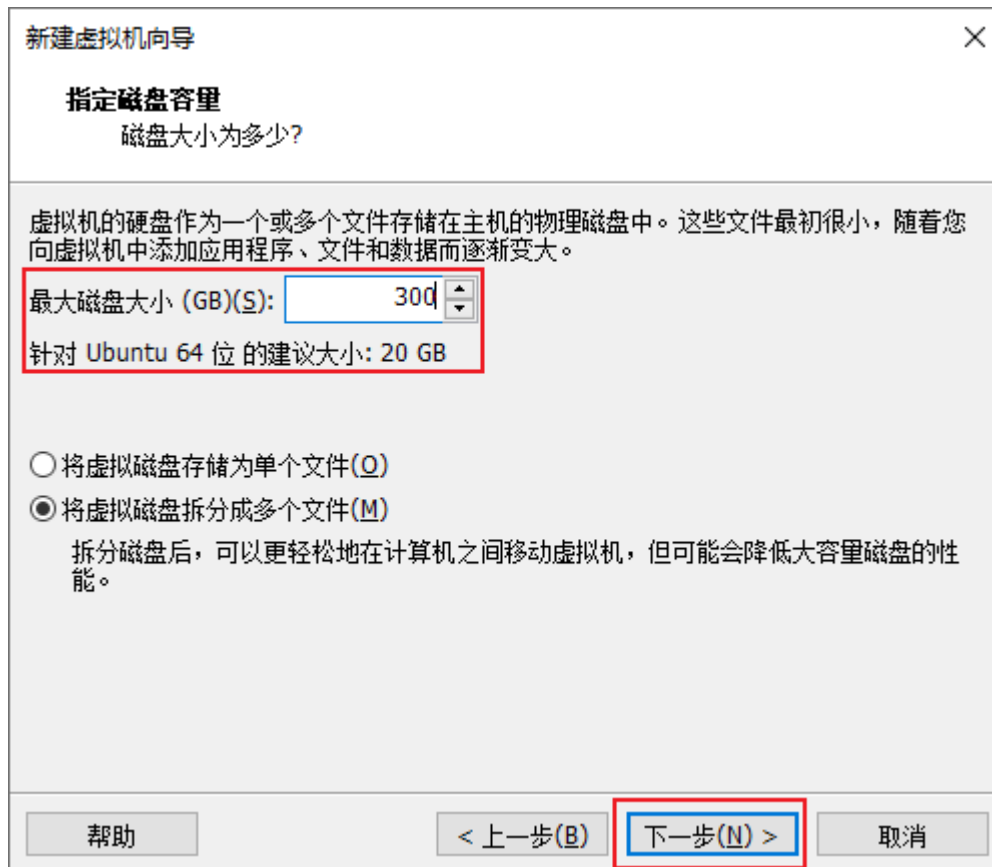
Set the VM name and storage location; a disk with plenty of free space is recommended.



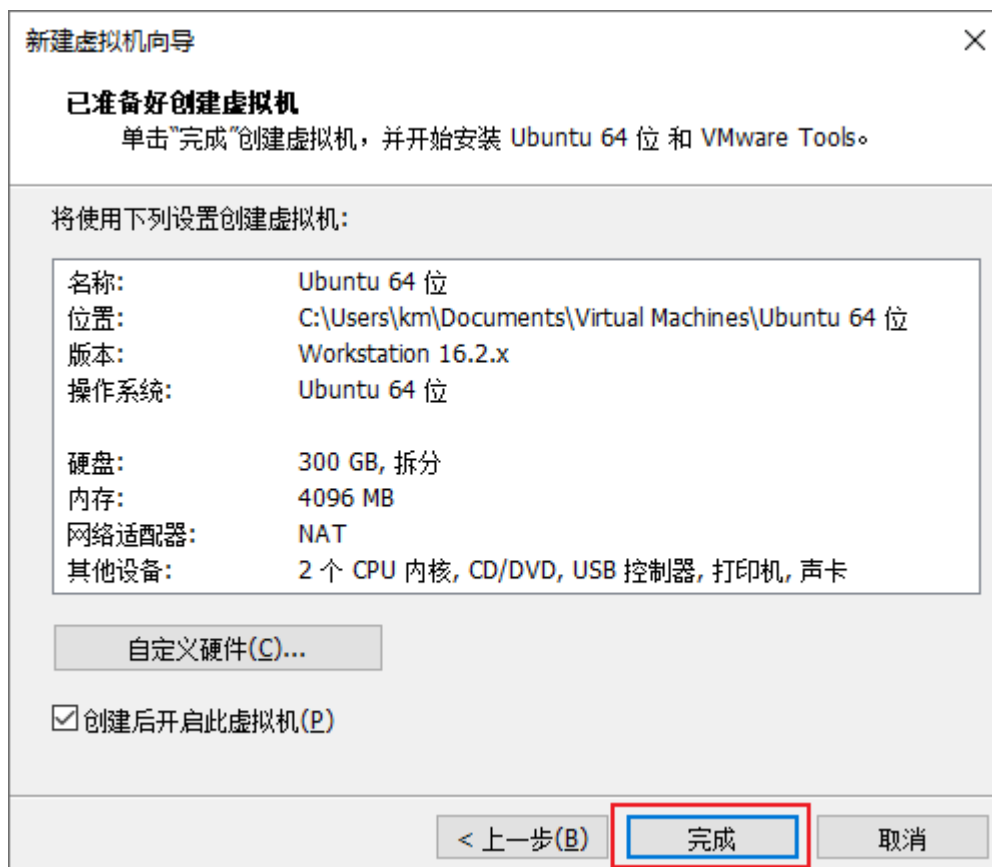
Set the disk capacity to 60GB or more, and choose "Split virtual disk into multiple files".



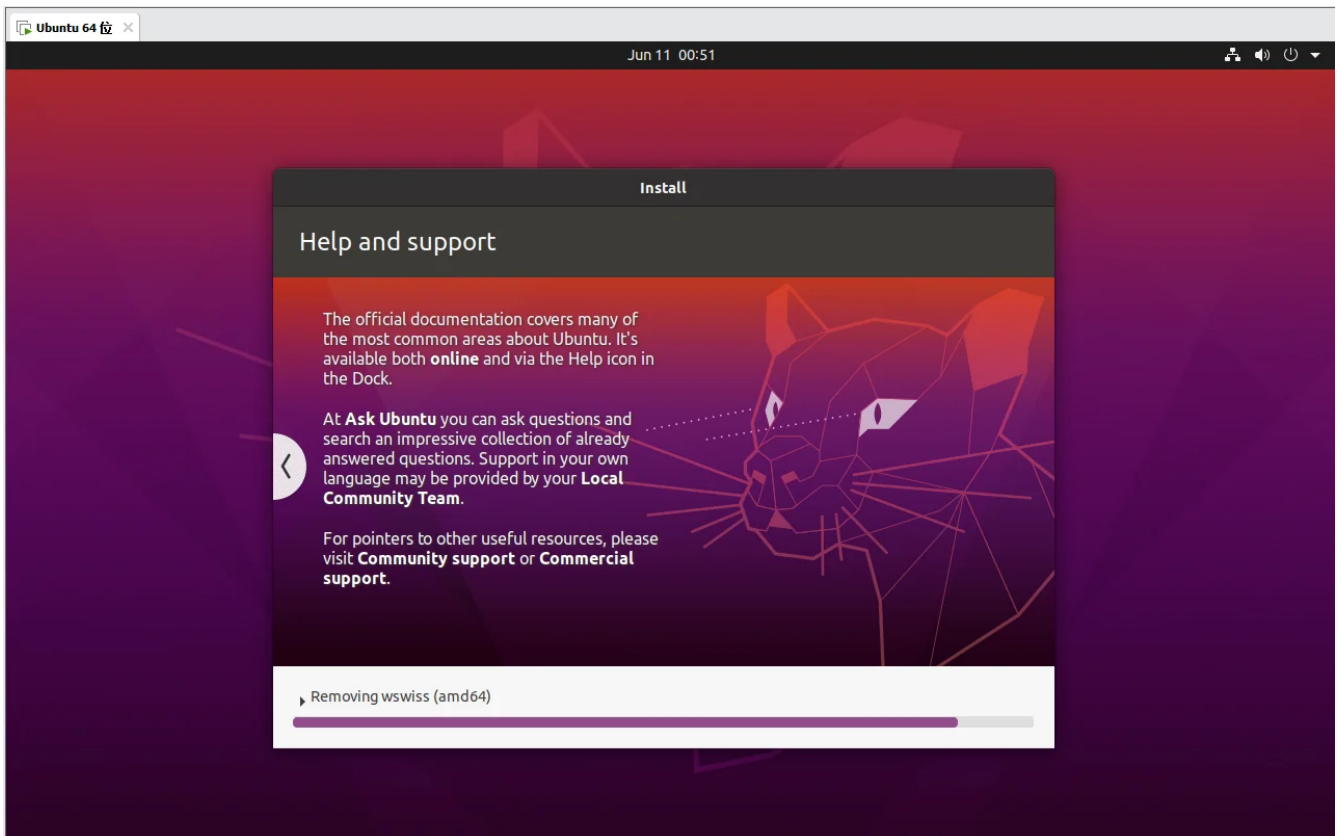
Click "Customize Hardware" to adjust memory, processors, and other settings.



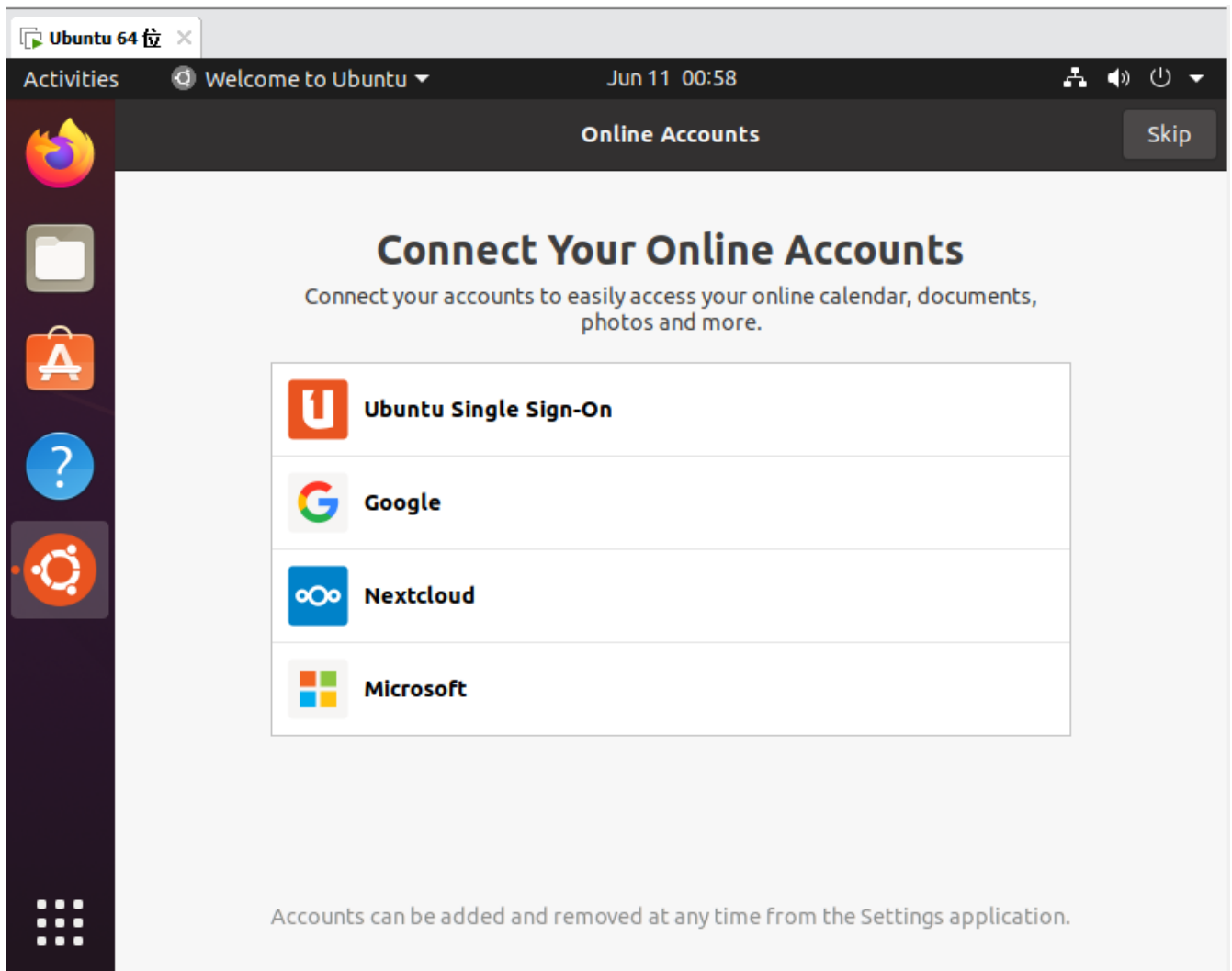
Set the memory to 8GB or more, and configure the processors based on your host.



When done, power on the virtual machine and begin the Ubuntu installation.



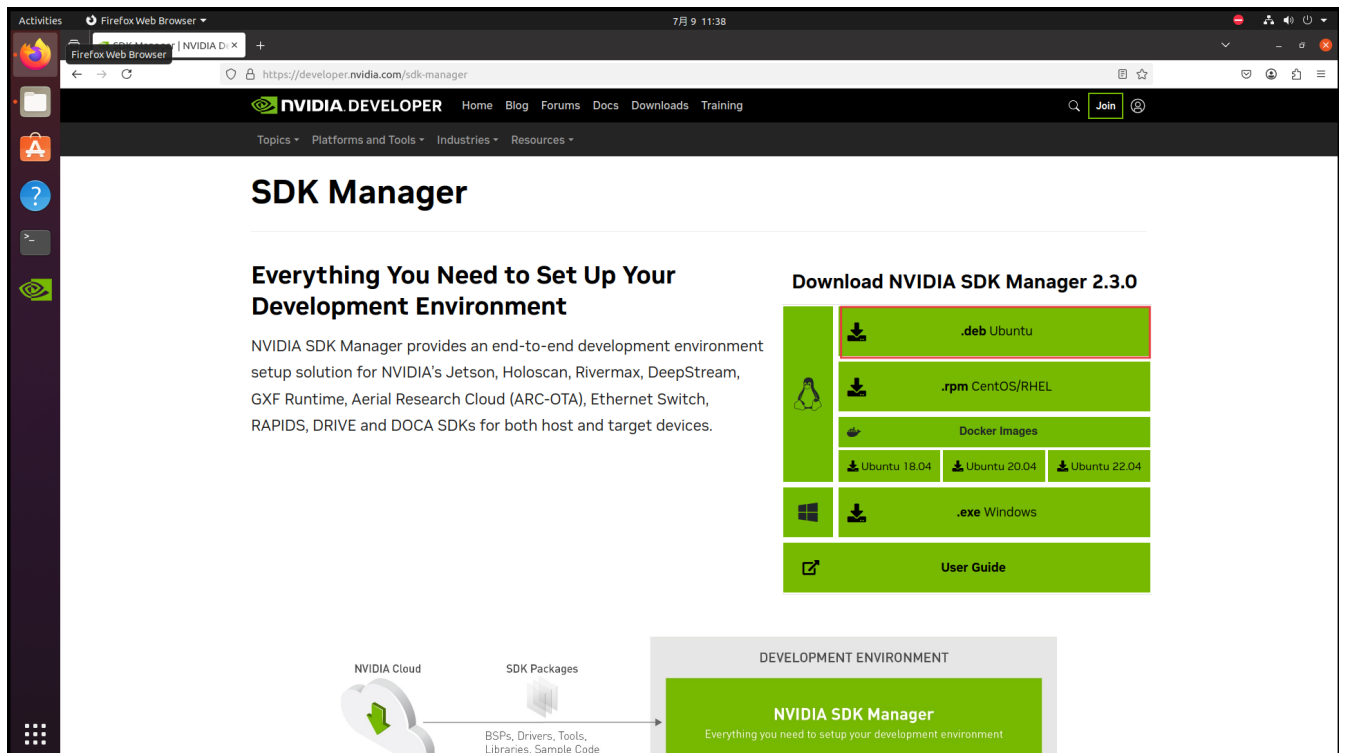
Follow the prompts to complete the Ubuntu installation.



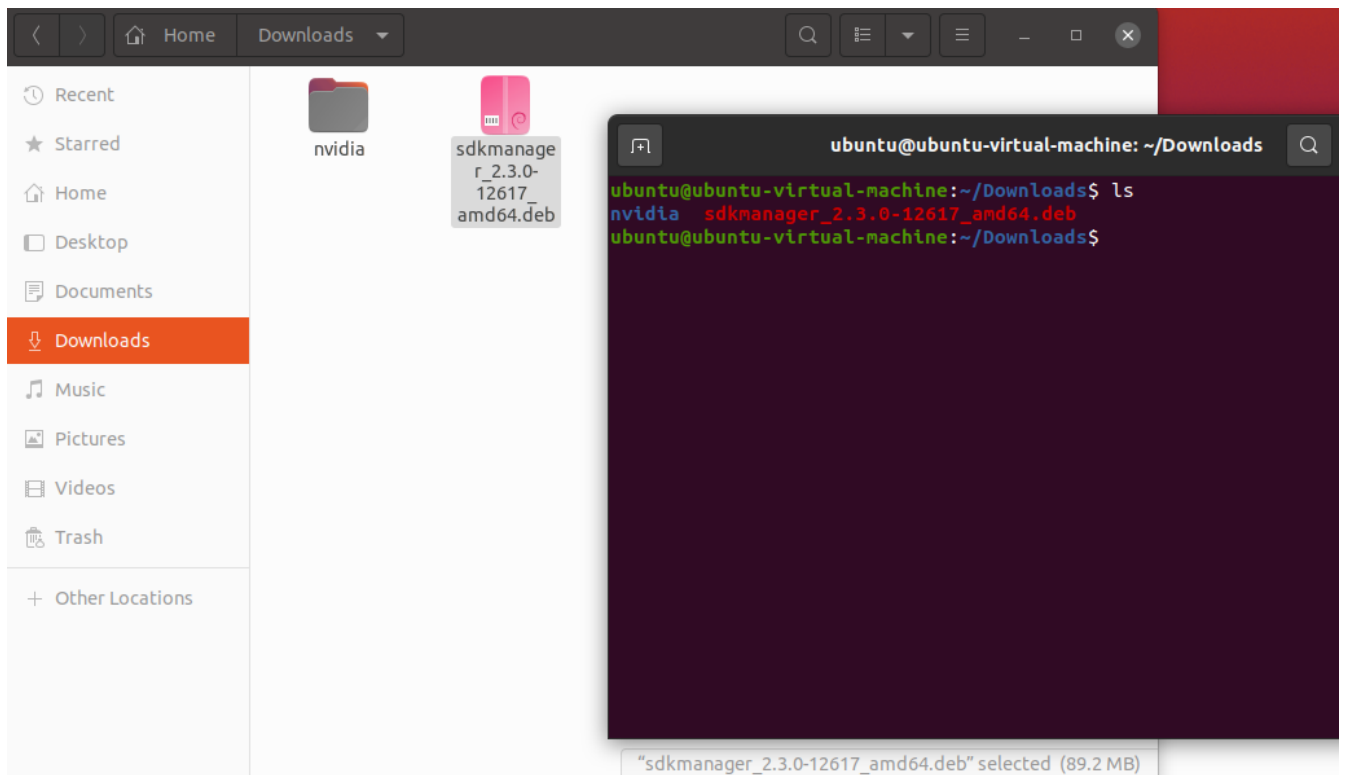
3. Install SDK Manager

Once inside Ubuntu, open a terminal and install the SDK Manager deb package you downloaded:

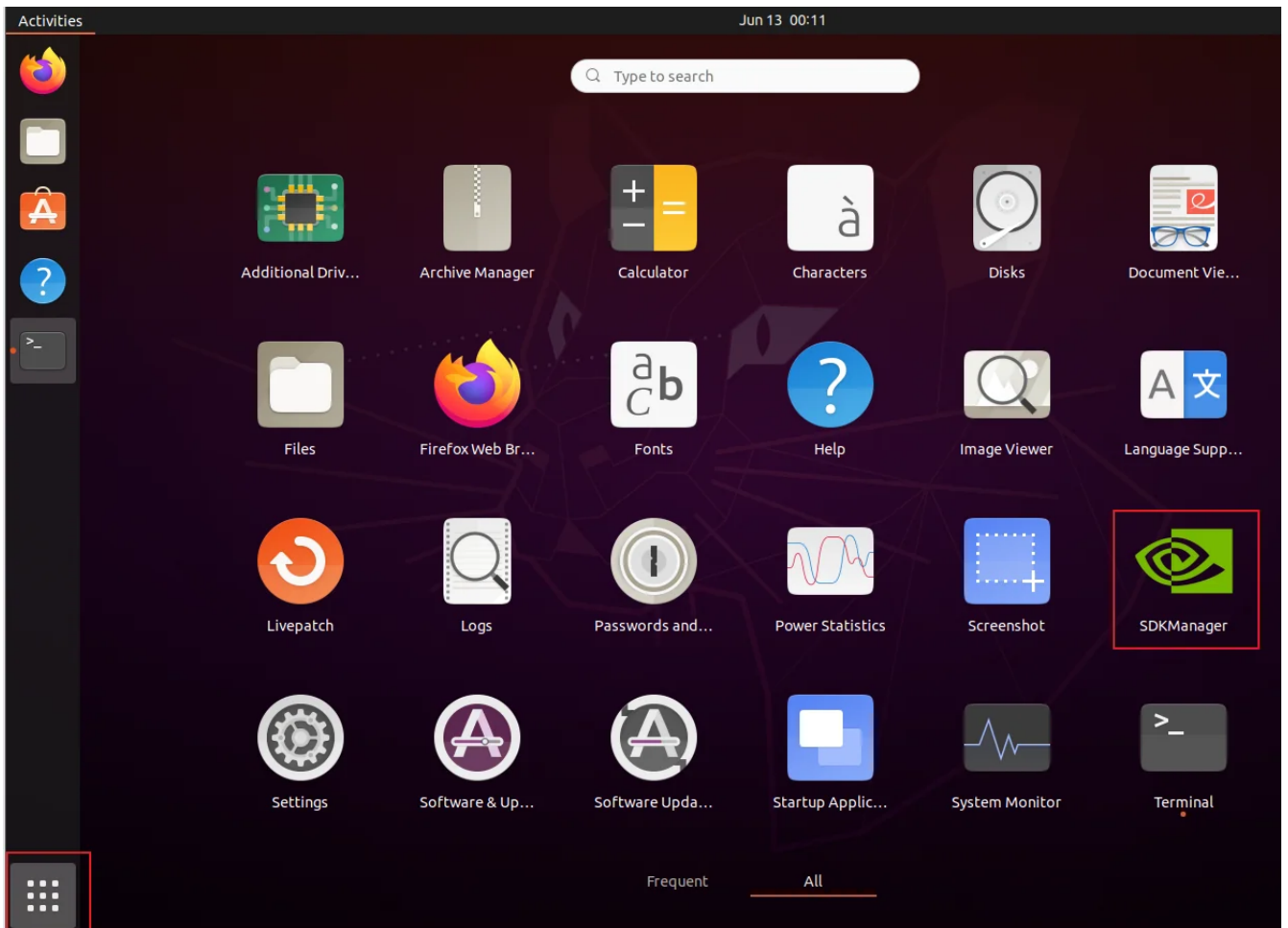
```
sudo apt install ./sdkmanager_*.deb
```



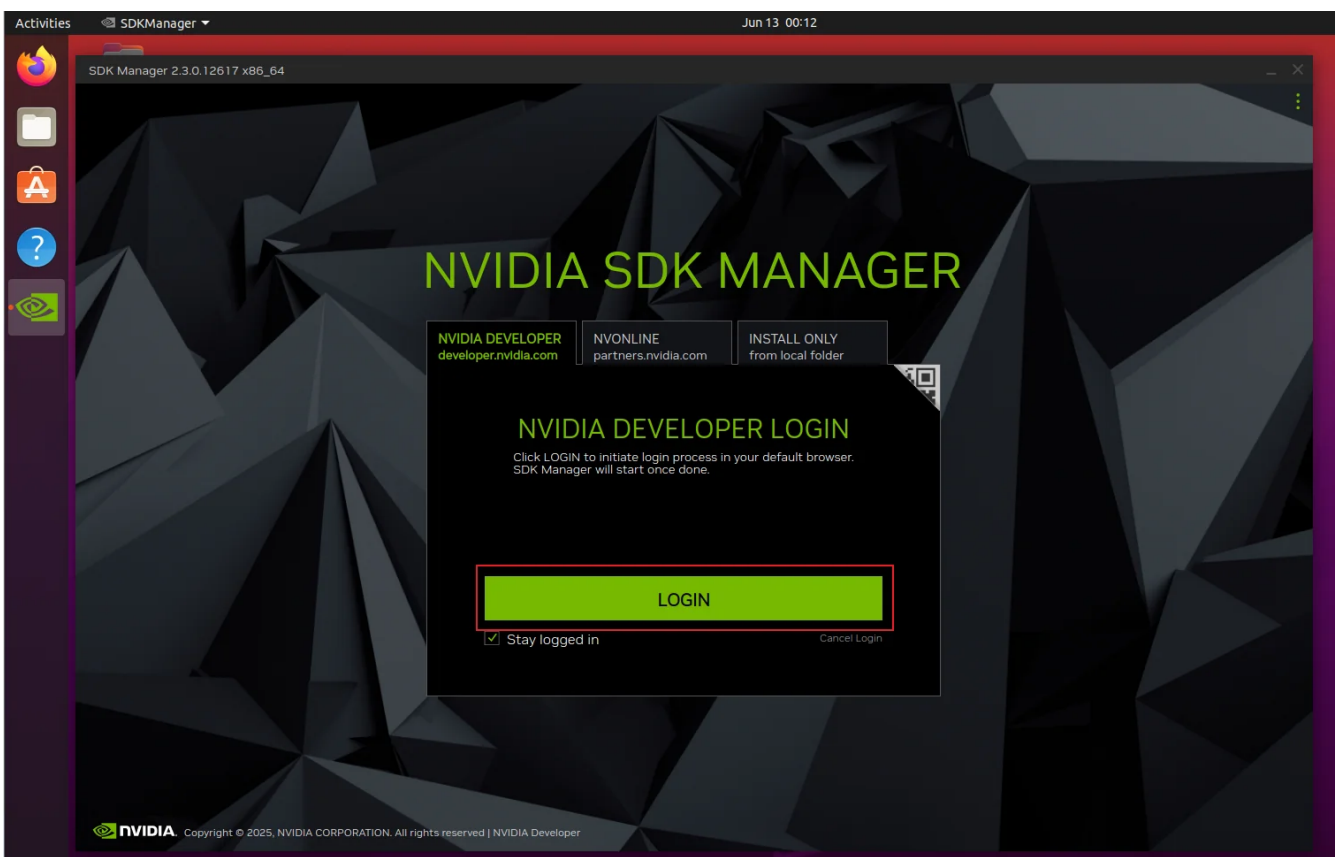
After installation, find and open SDK Manager from the application list.



Log in with your NVIDIA account (register in advance).



Once logged in, you will see the SDK Manager main screen, ready to start flashing.



Next Steps

- [Flash the Official Developer Kit](#)
- [Flash the C1901](#)
- [Flash the C1902](#)
- [Flash the C2401](#)

Source: [Kytech \(Guangzhou\) Co., Ltd](#) — LinkZee Labs

Flash the Official Developer Kit

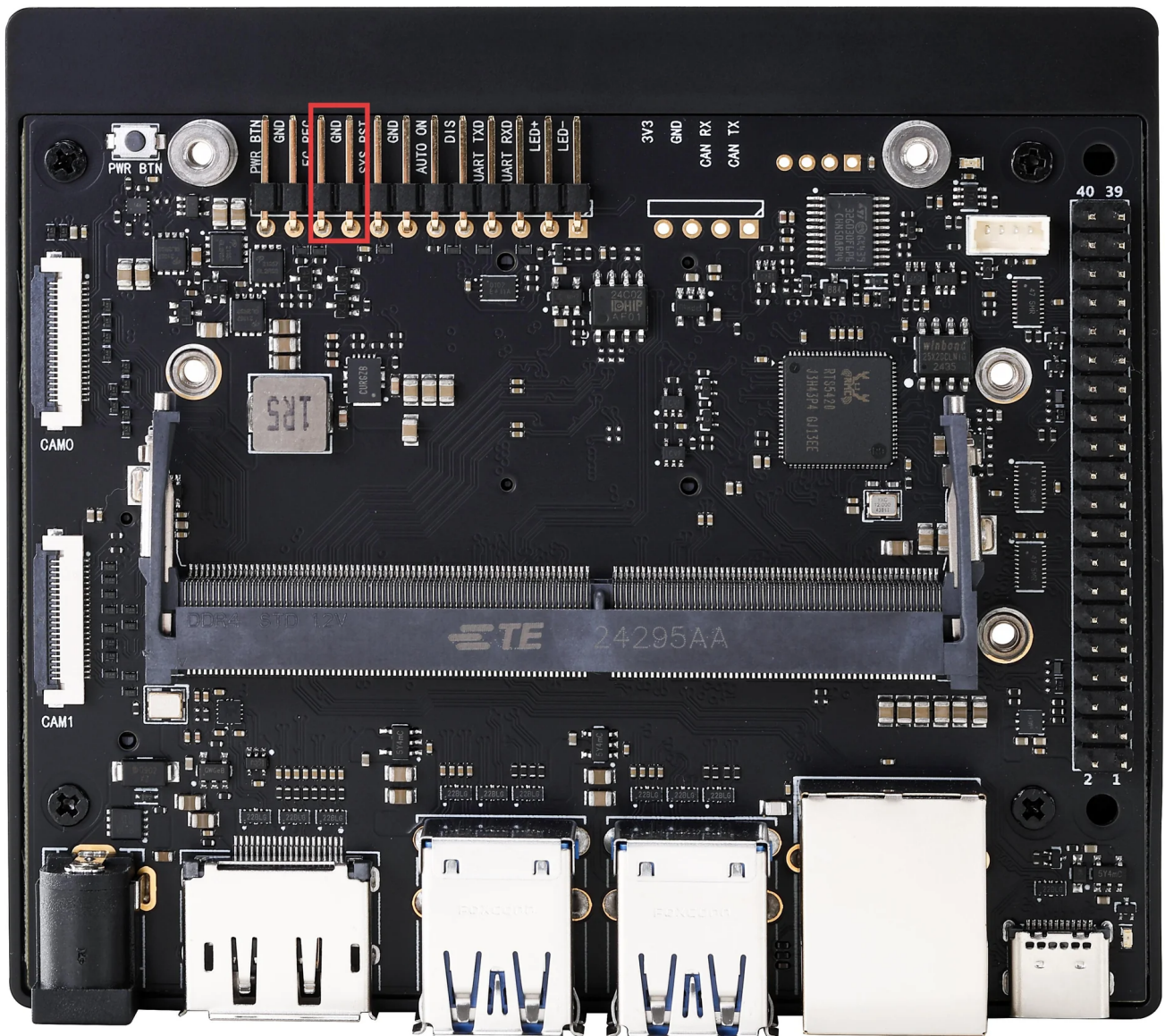
This guide explains how to use **SDK Manager** to flash the system onto an **NVIDIA official Developer Kit**.

Scope

This tutorial applies to the **Jetson Orin Nano/NX official Developer Kit**. For third-party carrier boards (C1901/C1902/C2401), refer to the corresponding flashing guides.

1. Enter Recovery Mode

Before flashing, put the module into **Recovery Mode**.



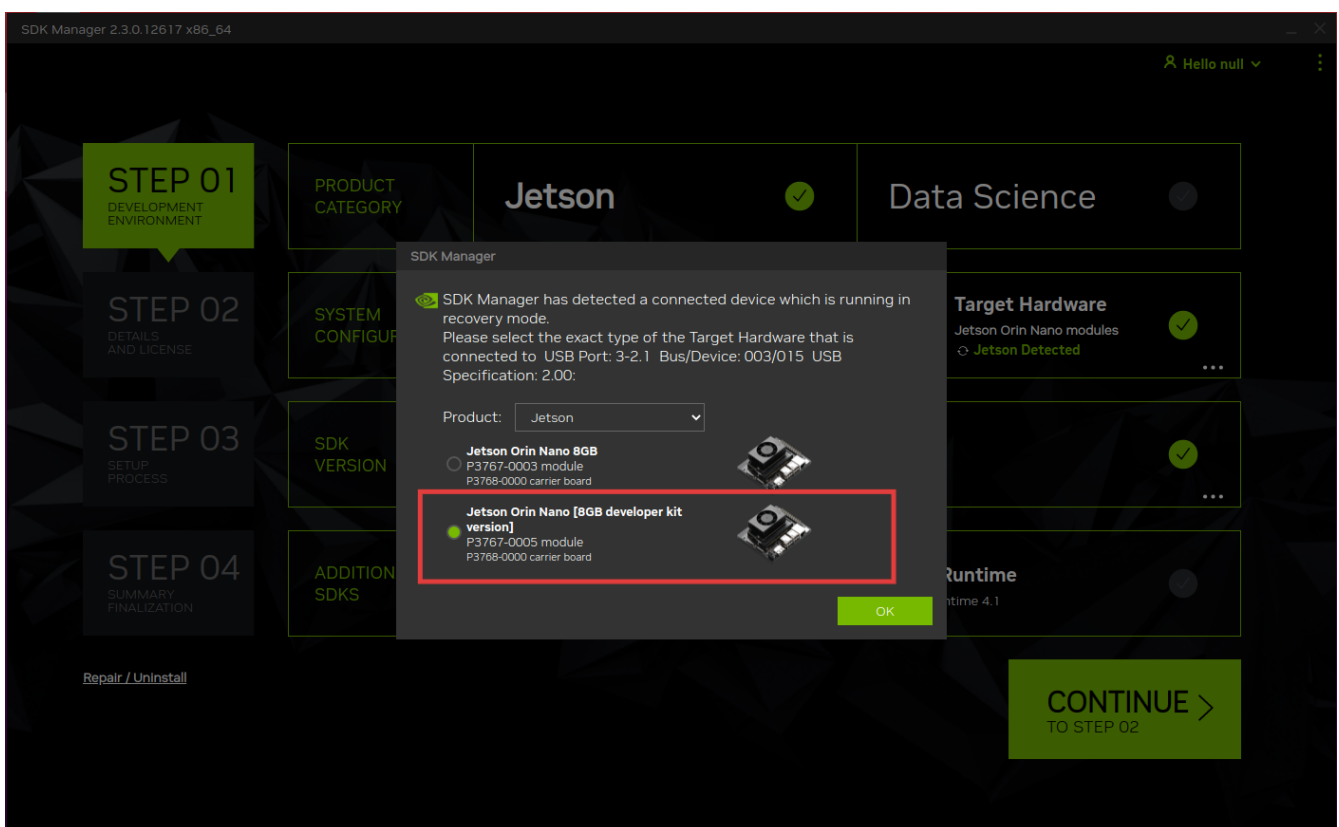
Steps:

1. Disconnect the power.
2. Use a jumper cap to short the **FC REC** and **GND** pins (or hold the Recovery button).
3. Connect the power, hold for 2 seconds, then release.
4. Connect the module to the virtual machine with a USB cable.

2. Detect the Device

Run the following command in the Ubuntu terminal to confirm the device is detected:

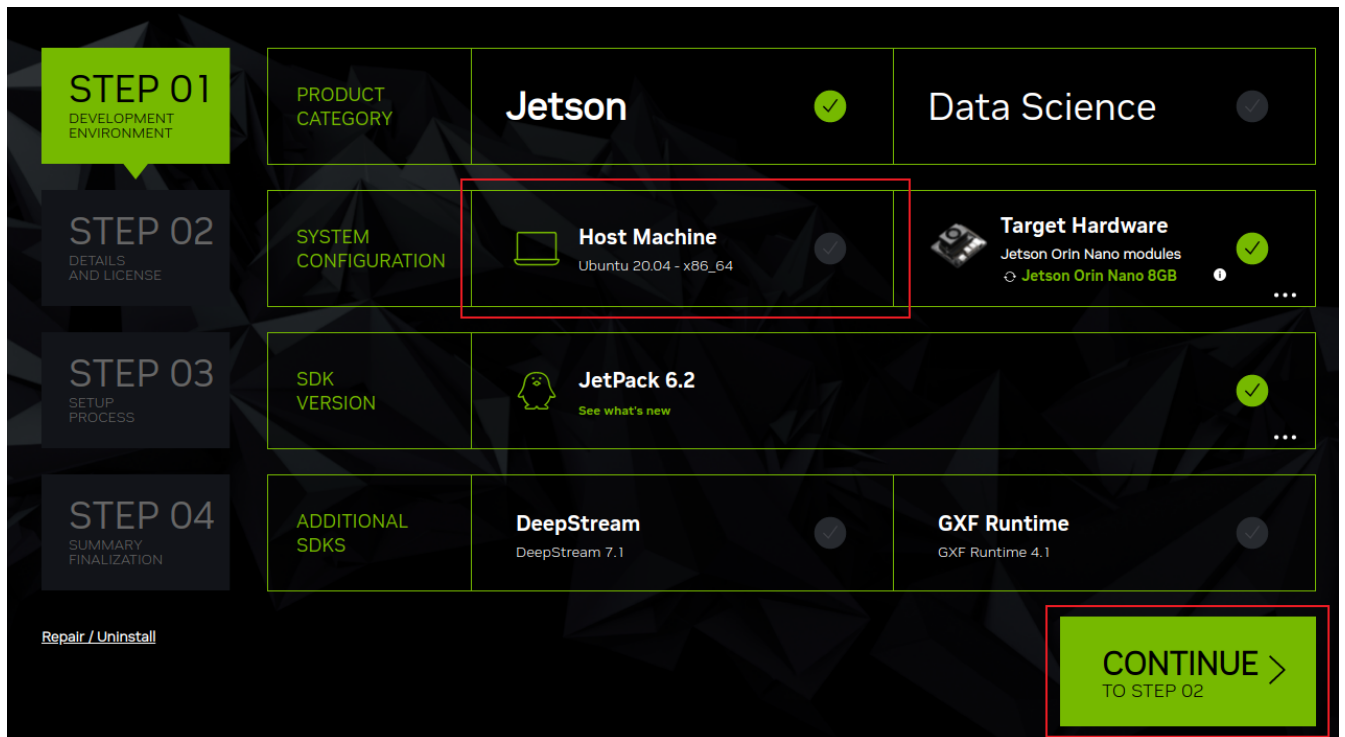
```
lsusb | grep -i nvidia
```



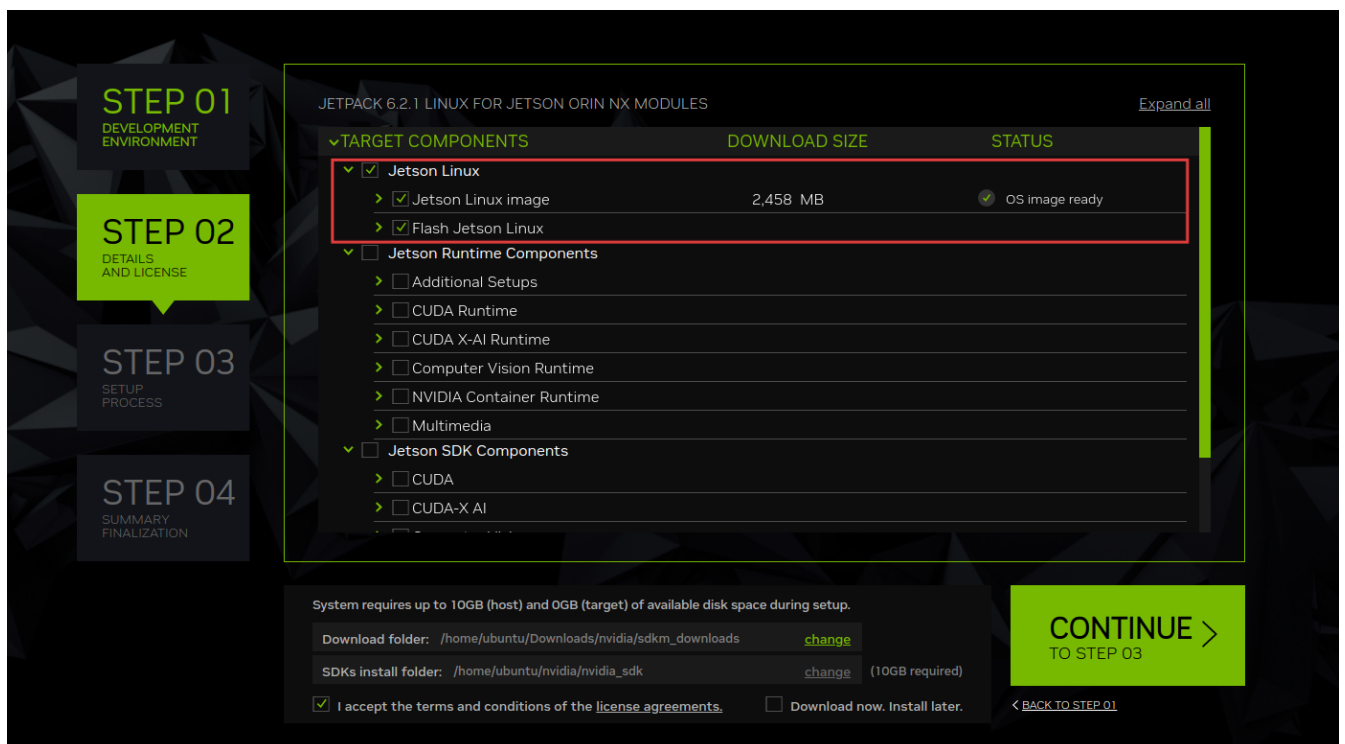
If you see NVIDIA-related devices, recovery mode was entered successfully and the VM has captured the USB device.

3. Flash with SDK Manager

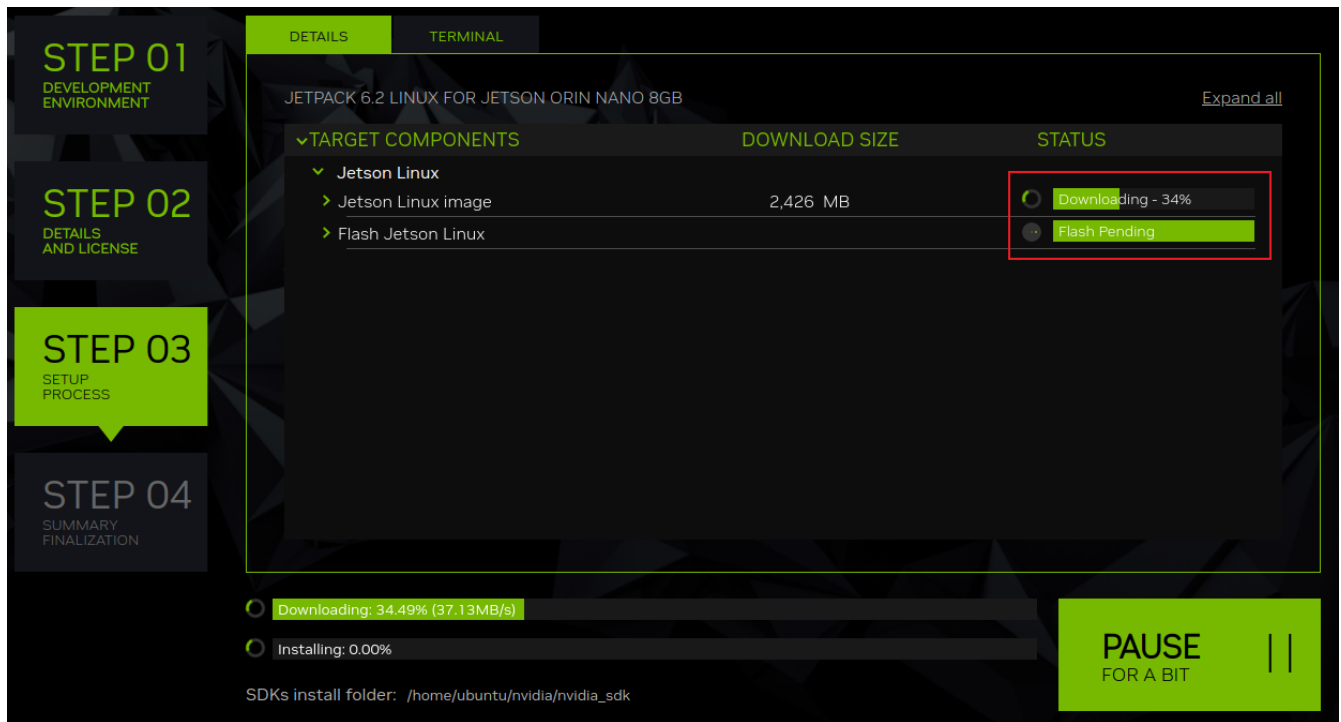
Open SDK Manager, log in, and select the corresponding hardware model.



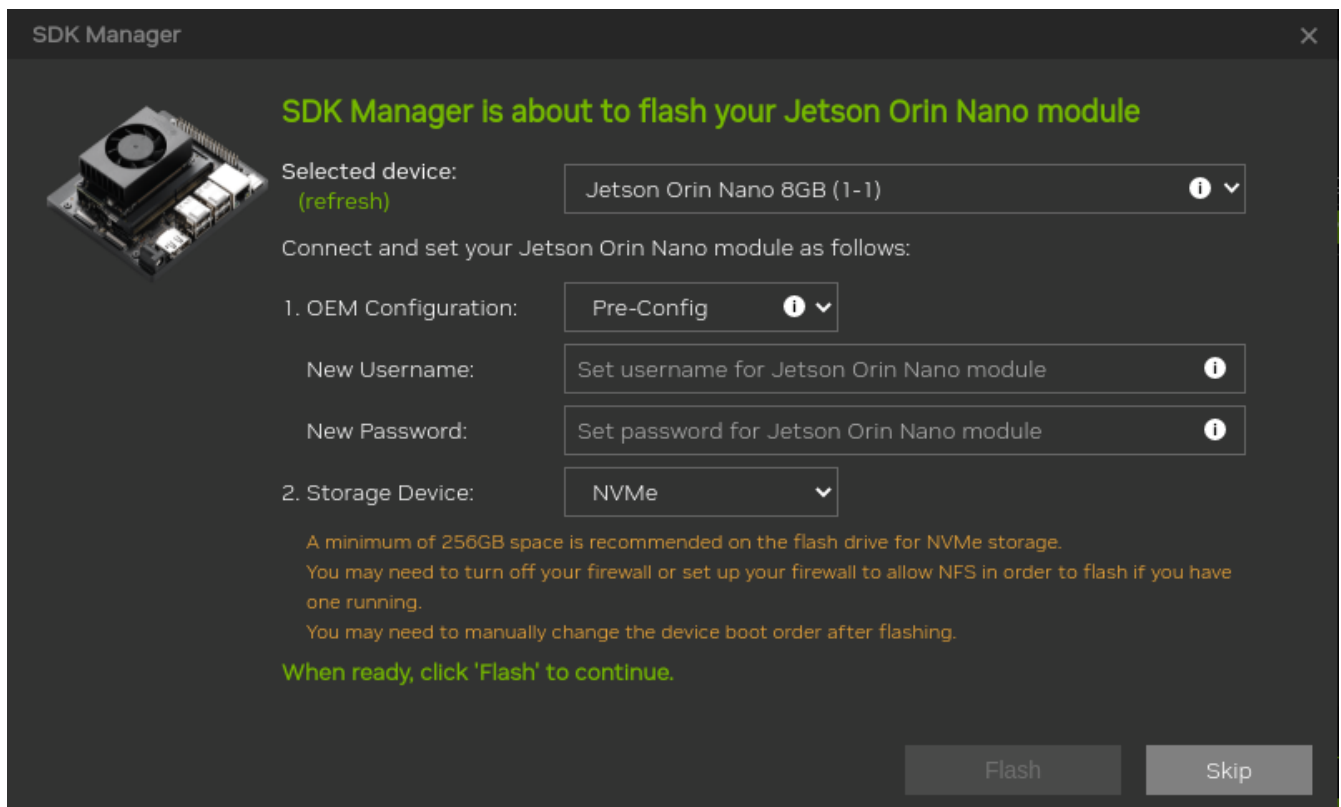
Uncheck **Host Machine** and keep only **Target Hardware**.



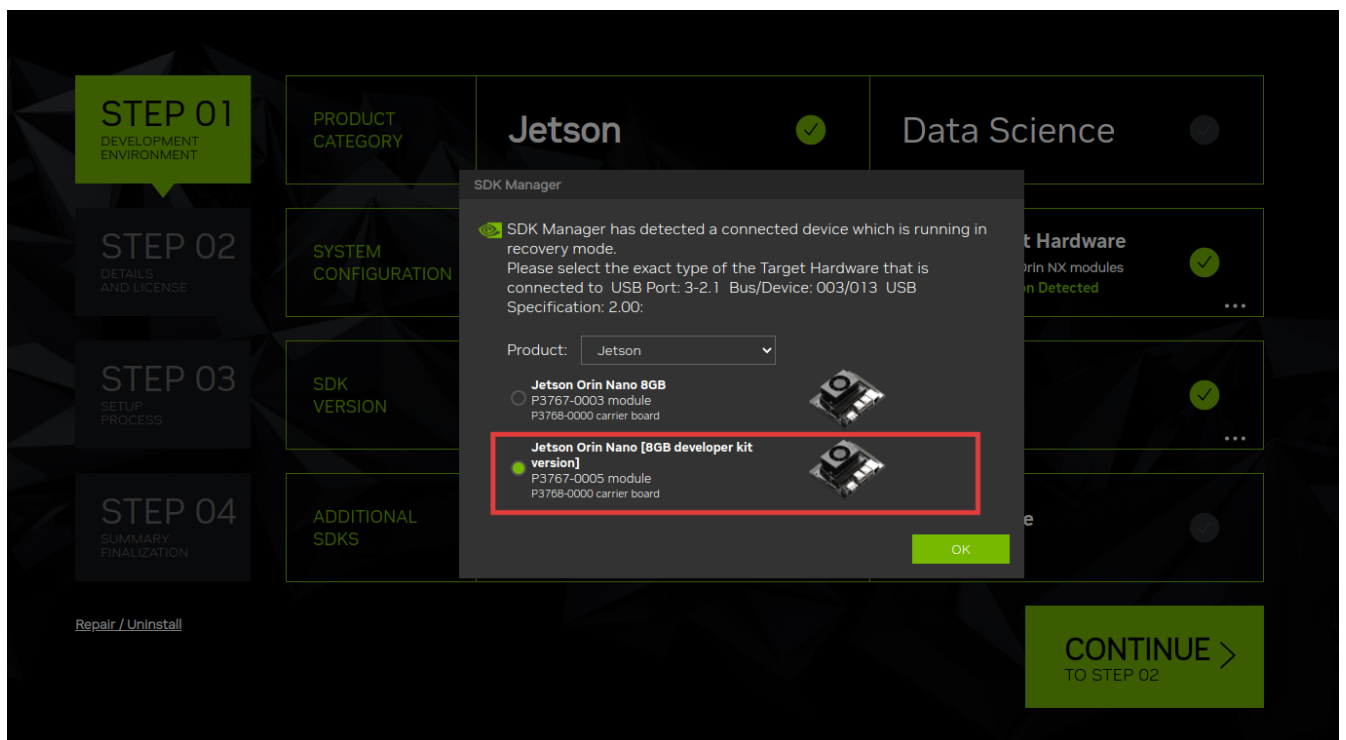
Select the JetPack version; the latest stable release is recommended.



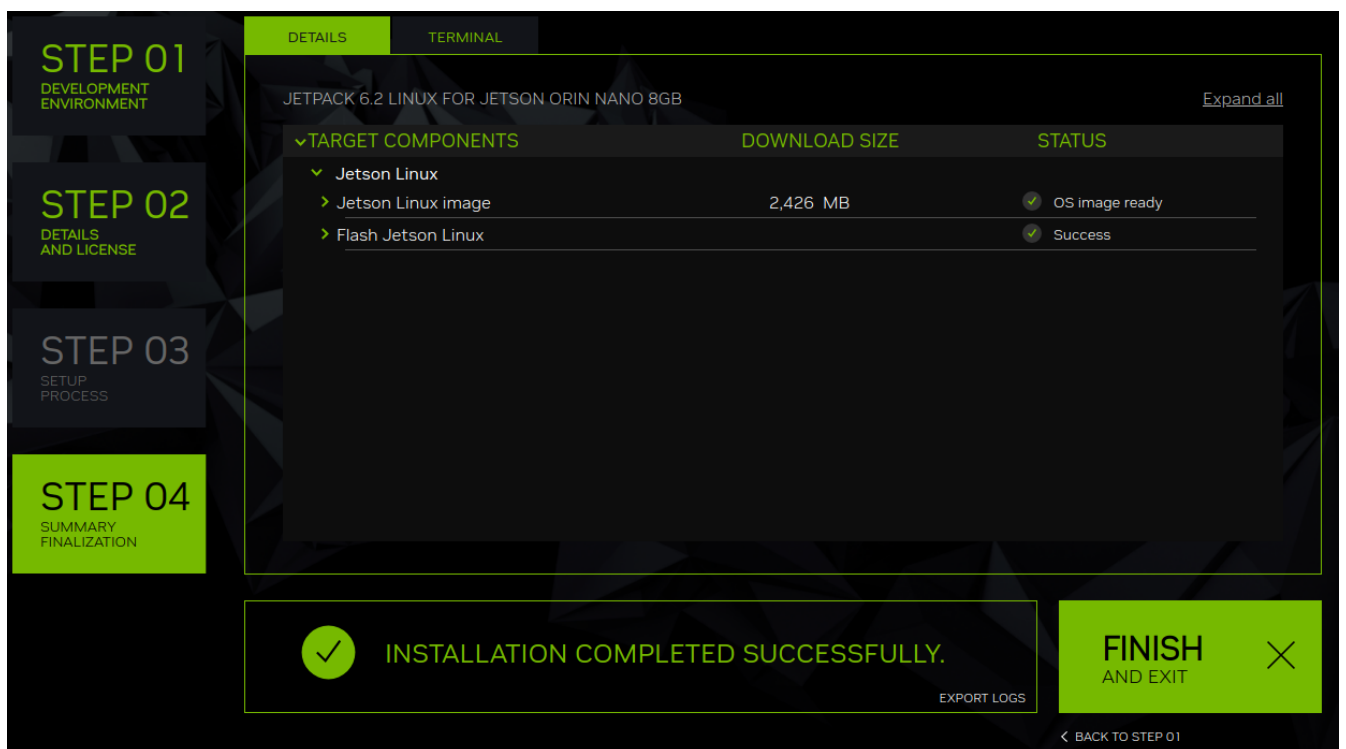
Proceed to the flashing flow, accept the license agreement, and start downloading and flashing the system.



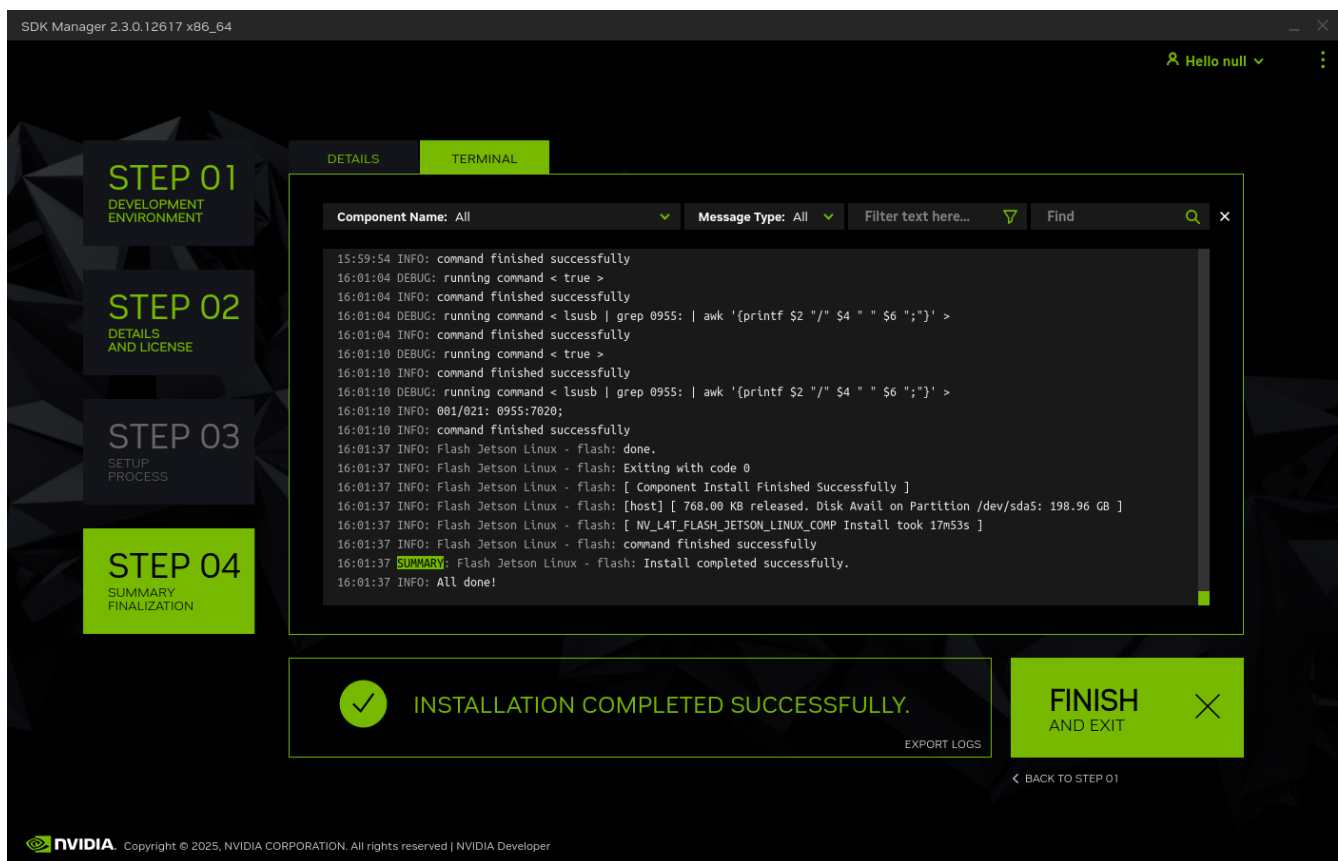
In the flashing configuration window that pops up, choose the manual setup mode.



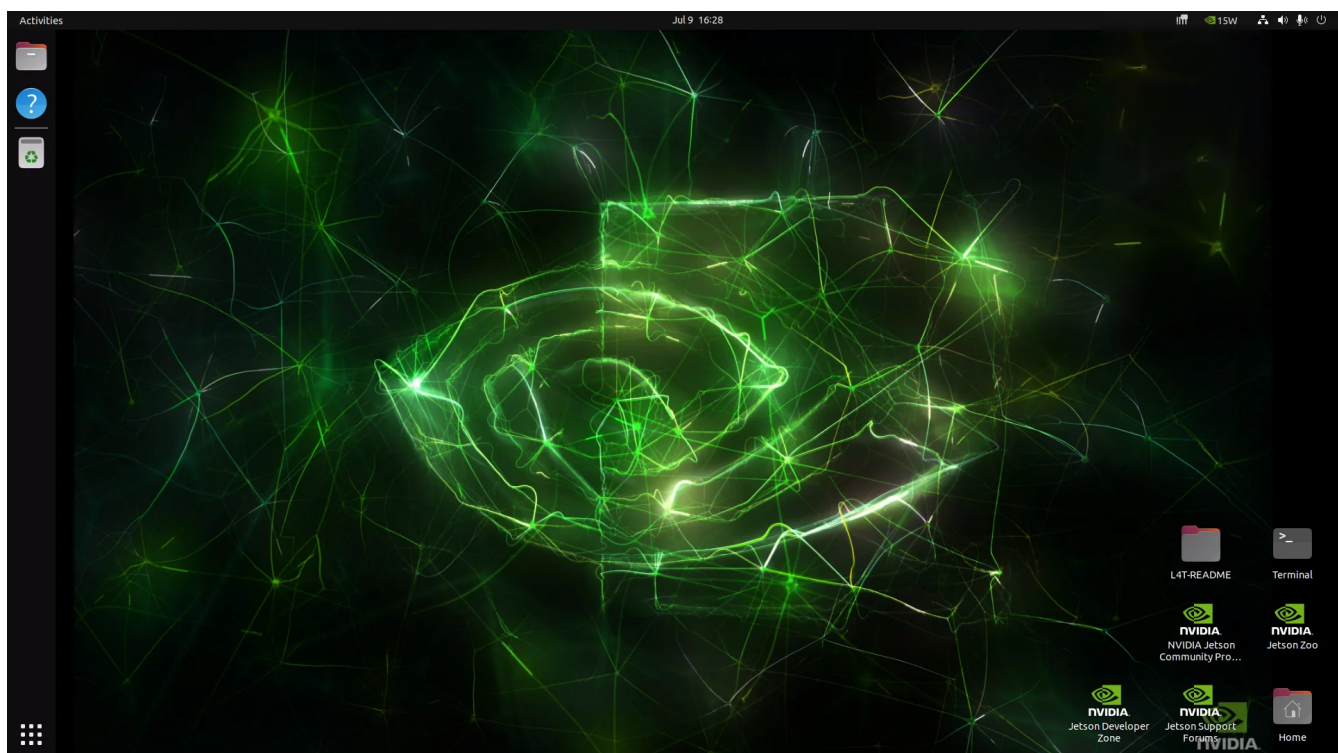
Choose the storage medium (such as NVMe or eMMC) and set a username and password.



Click Flash to start flashing the system and wait for it to finish.



After flashing, the module reboots automatically into the system.



Troubleshooting

Device Not Detected

- Confirm recovery mode was entered correctly
- Check that the USB cable is a data cable (not charge-only)
- In VMware, confirm the USB device is connected to the VM (VM > Removable Devices)
- Try a different USB port or cable

Flashing Interrupted or Failed

- Ensure the VM has enough disk space ($\geq 60\text{GB}$)
- Disable the VM's power-saving/sleep settings
- Do not disconnect USB during flashing
- Re-enter recovery mode and retry

Choosing a Storage Medium

- **NVMe SSD**: Fast and high-capacity — recommended
- **eMMC**: Onboard storage, fixed capacity
- **SD card**: Supported only on some models

Next Steps

- [Flash the C1901](#)
- [Flash the C1902](#)
- [Flash the C2401](#)

Source: [Kytech \(Guangzhou\) Co., Ltd](#) — [LinkZee Labs](#)

Flash the C1901

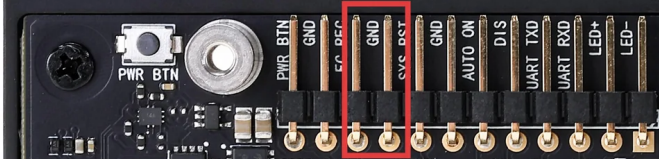
This guide explains how to flash the system onto the **C1901 carrier board**. C1901 is developed from the reference design, so the flashing process is essentially the same as the official Developer Kit.

Before You Start

- Ubuntu VM and SDK Manager installed (see the [installation guide](#))
- A USB Type-C cable capable of data transfer
- C1901 carrier board + Jetson Orin Nano/NX module

1. Enter Recovery Mode

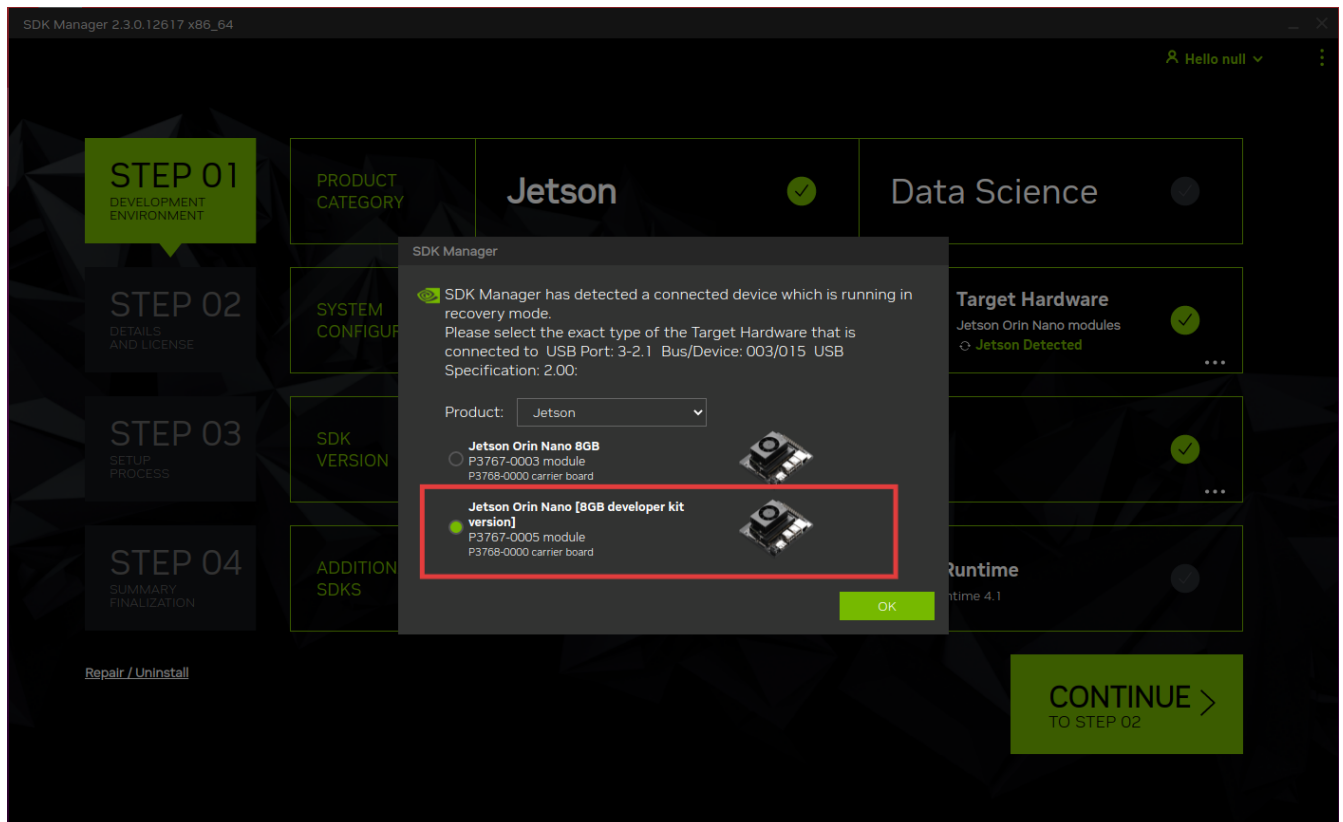
The C1901 enters recovery mode through the **12-pin debug header**.



1. Disconnect the C1901 power.
2. Use a jumper cap to short the **FC REC** and **GND** pins.
3. Connect the power.
4. Connect the C1901 to the VM with a USB Type-C cable.
5. Remove the jumper cap.

Run in the Ubuntu terminal:

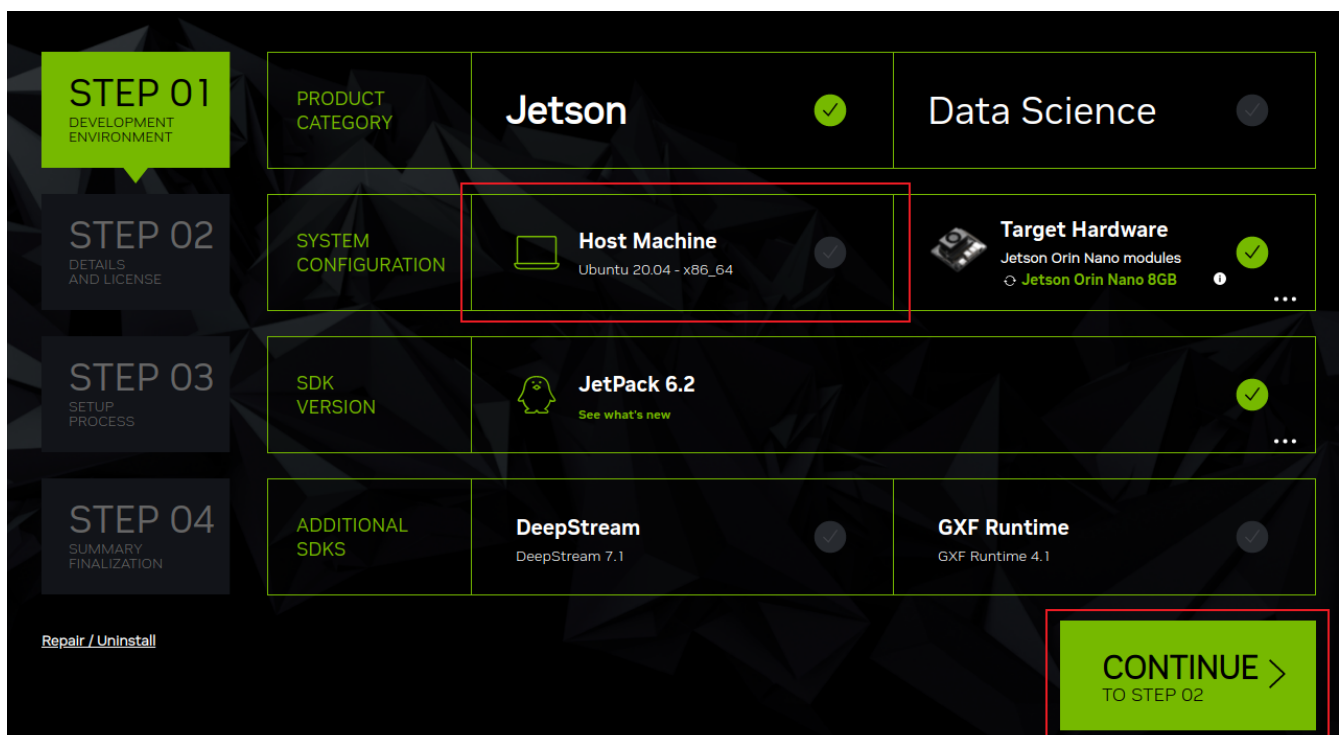
```
lsusb | grep -i nvidia
```



If an NVIDIA device is detected, recovery mode was entered successfully.

3. Flash with SDK Manager

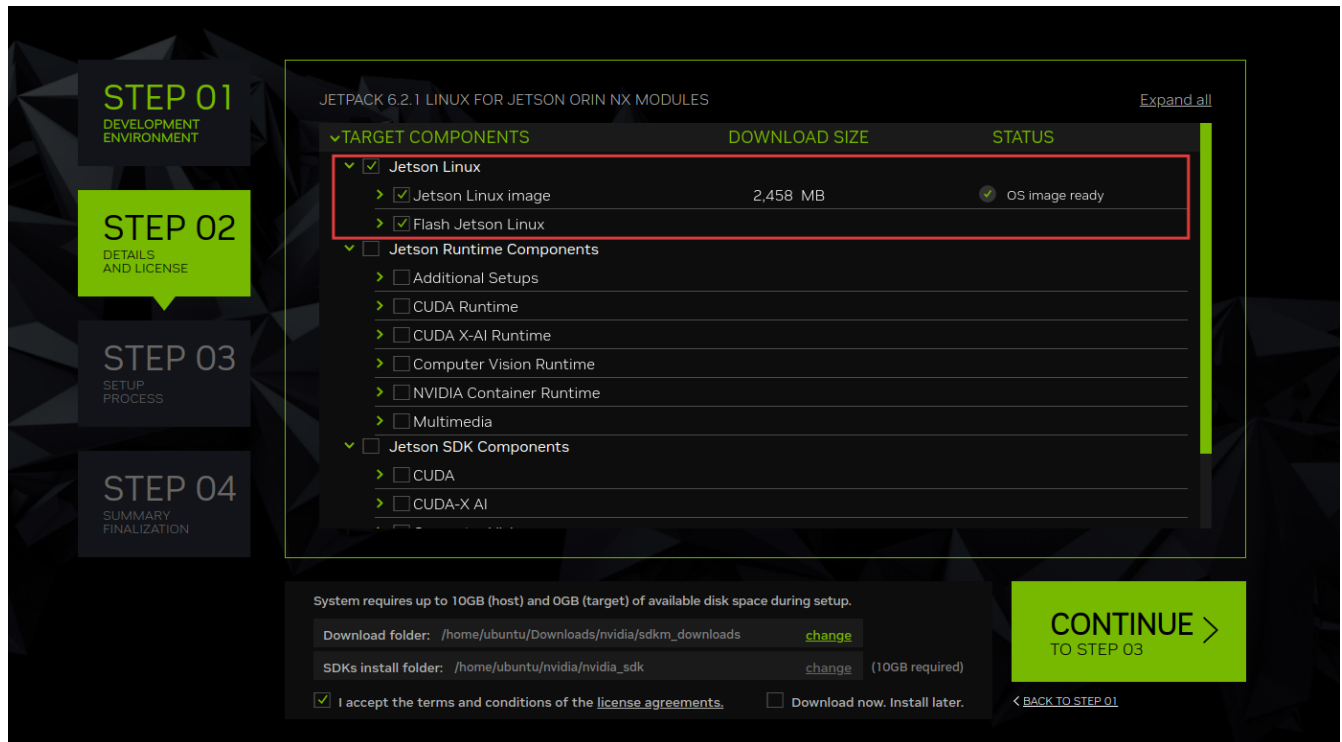
Open SDK Manager and select the corresponding module model (Orin Nano or Orin NX).



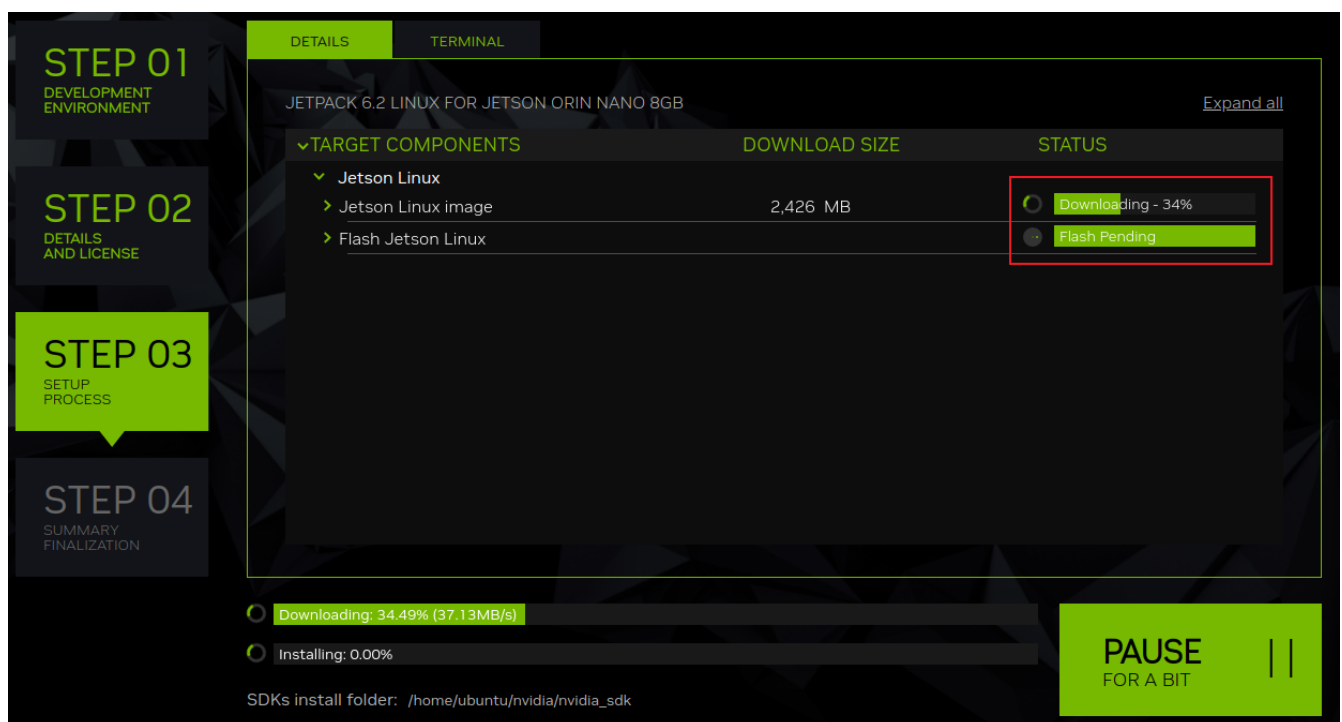
Note

For the C1901, select the model that matches your module. Do not choose custom options other than the Developer Kit version.

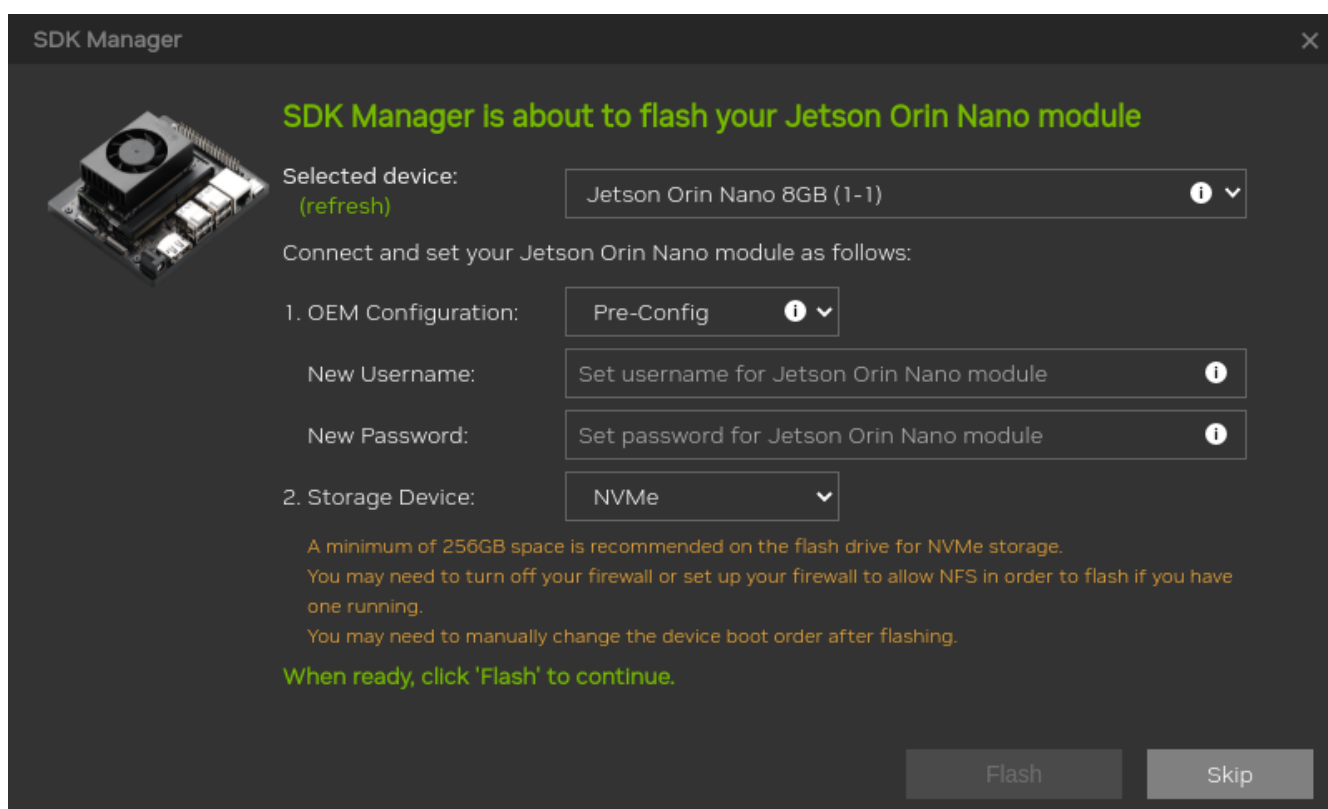
Uncheck Host Machine and keep only Target Hardware.



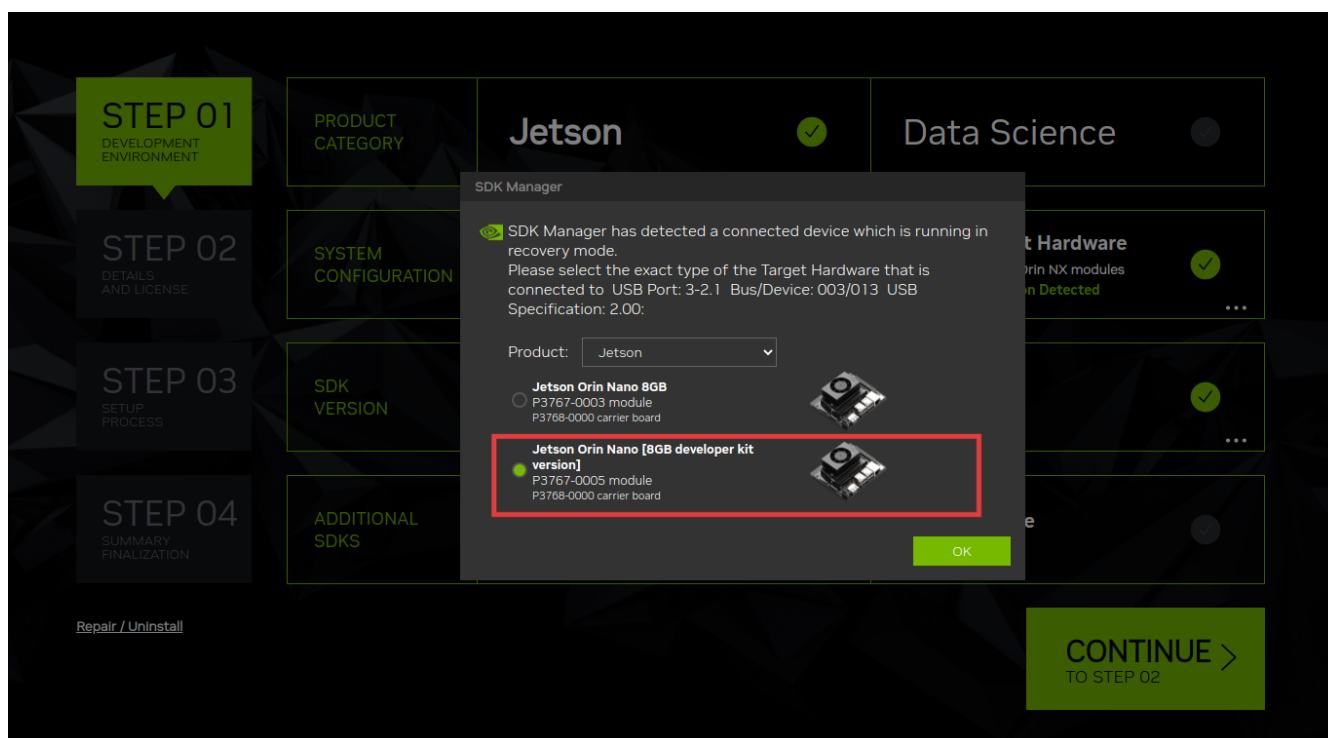
After selecting the JetPack version, start the flashing flow.



In the flashing configuration window, select the storage medium and system version.



Click Flash to start and wait for it to finish.



4. Finish

After flashing, the C1901 reboots automatically into the system. The first boot requires initial system configuration.



First Boot Configuration

- Select language, time zone, and keyboard layout
- Set a username and password
- Once done, you reach the Ubuntu desktop

Troubleshooting

Cannot Enter Recovery Mode

- Confirm the FC REC and GND pins are shorted correctly
- Confirm the power supply is working
- Check that the USB Type-C cable is a data cable

SDK Manager Cannot Detect the Device

- In VMware, confirm the USB device is connected to the VM
- Run `lsusb` to confirm the device is detected
- Try a different USB port

Flashing Failed

- Ensure enough disk space
- Keep the USB connection stable during flashing
- Re-enter recovery mode and retry

Next Steps

- [C1901 Introduction](#)

Source: [Kytech \(Guangzhou\) Co., Ltd](#) — LinkZee Labs

Flash the C1902

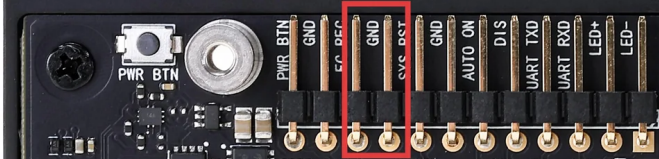
This guide explains how to flash the system onto the **C1902 carrier board**. C1902 is developed from the reference design and is fully compatible with the Developer Kit firmware.

Before You Start

- Ubuntu VM and SDK Manager installed (see the [installation guide](#))
- A USB Type-C cable capable of data transfer
- C1902 carrier board + Jetson Orin Nano/NX module

1. Enter Recovery Mode

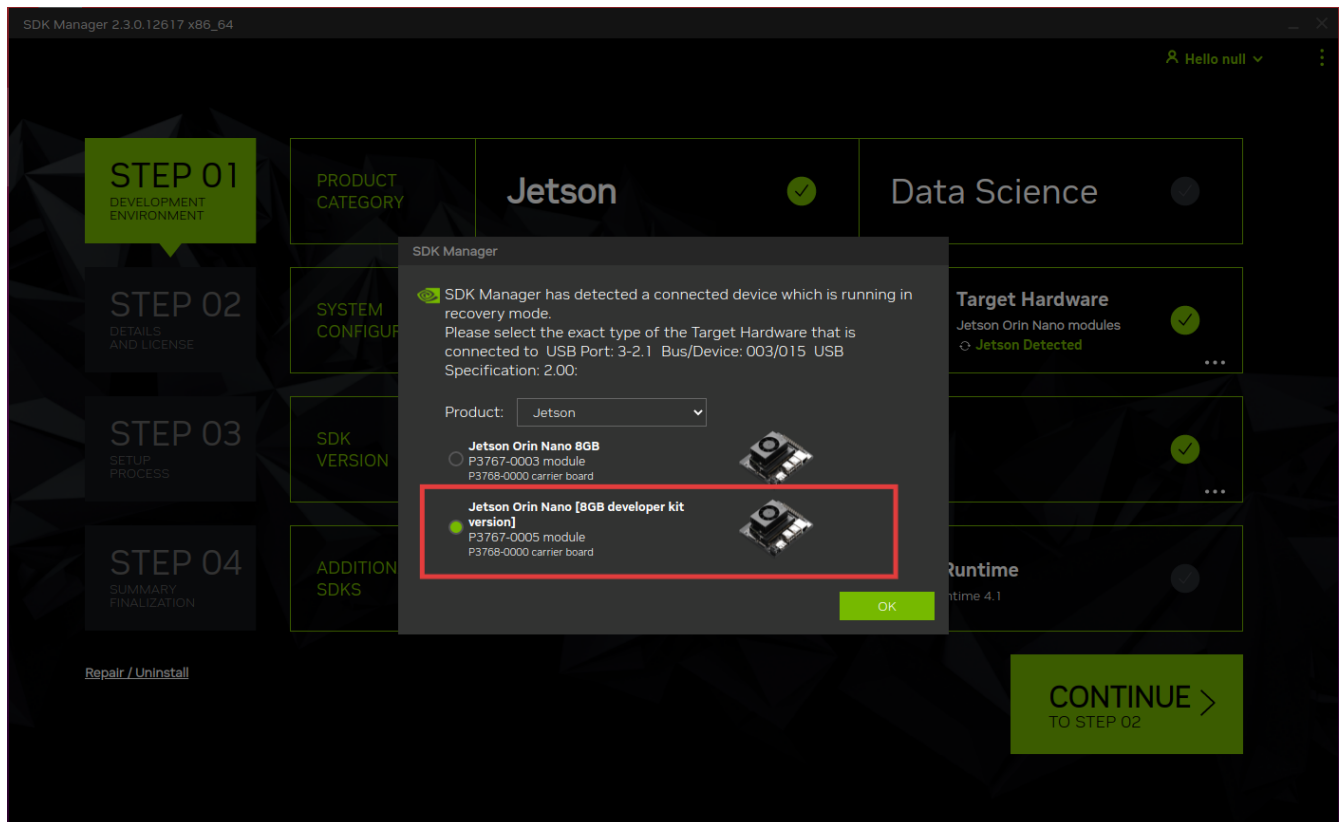
The C1902 enters recovery mode through the **12-pin debug header**.



1. Disconnect the C1902 power.
2. Use a jumper cap to short the **FC REC** and **GND** pins.
3. Connect the power.
4. Connect the C1902 to the VM with a USB Type-C cable.
5. Remove the jumper cap.

Run in the Ubuntu terminal:

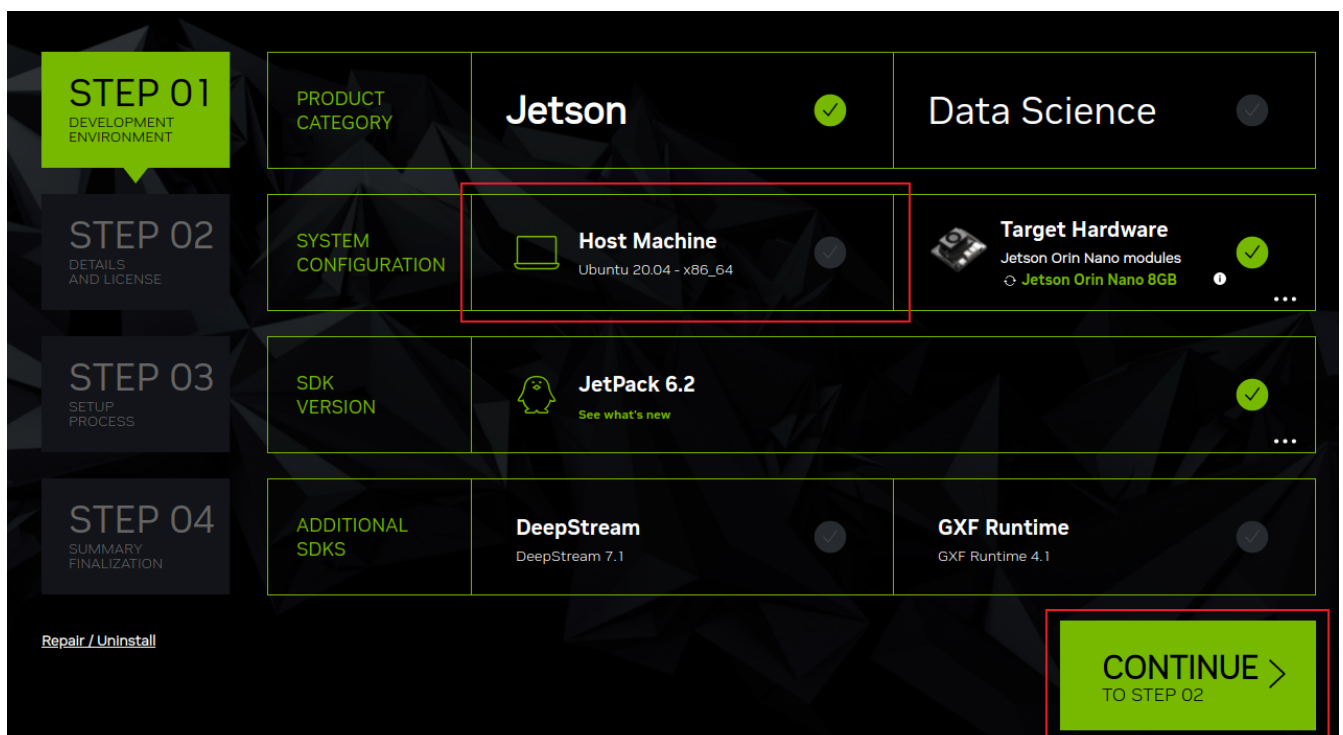
```
lsusb | grep -i nvidia
```

If an NVIDIA device is detected, recovery mode was entered successfully.

3. Flash with SDK Manager

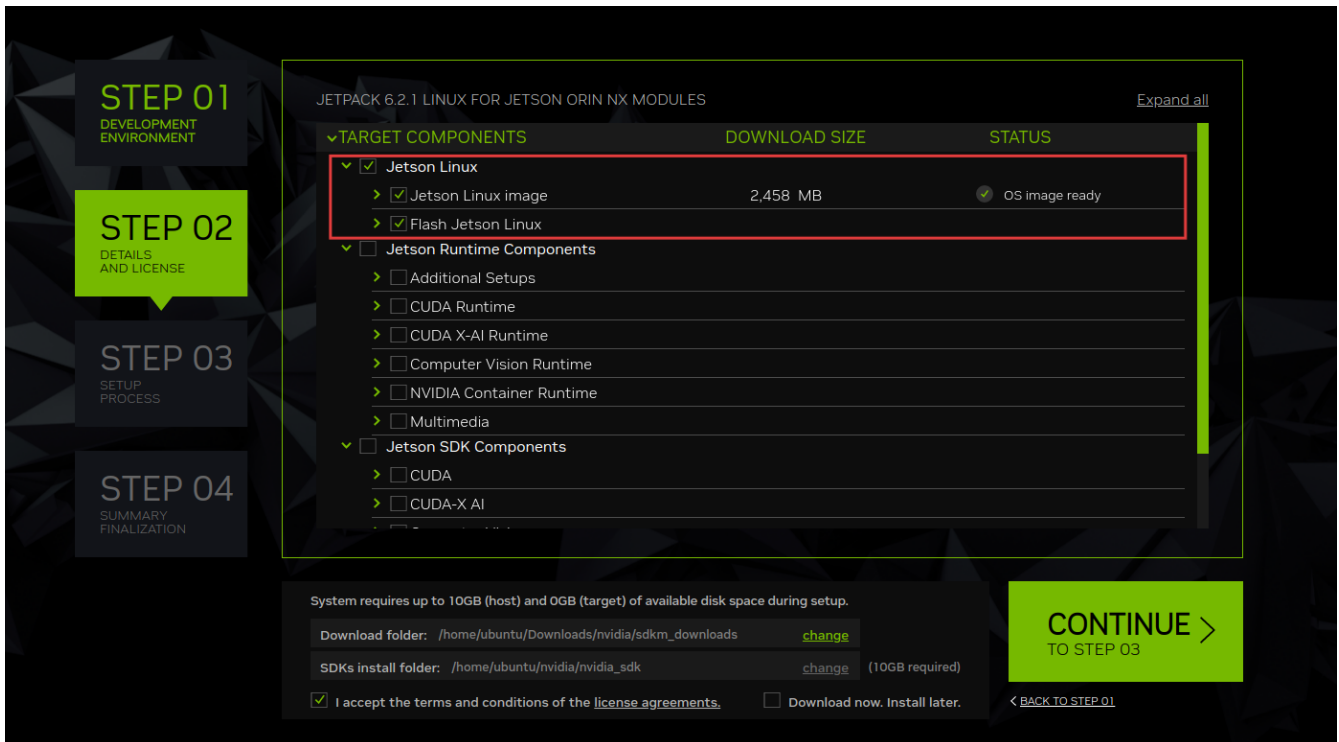
Open SDK Manager and select the corresponding module model (Orin Nano or Orin NX).



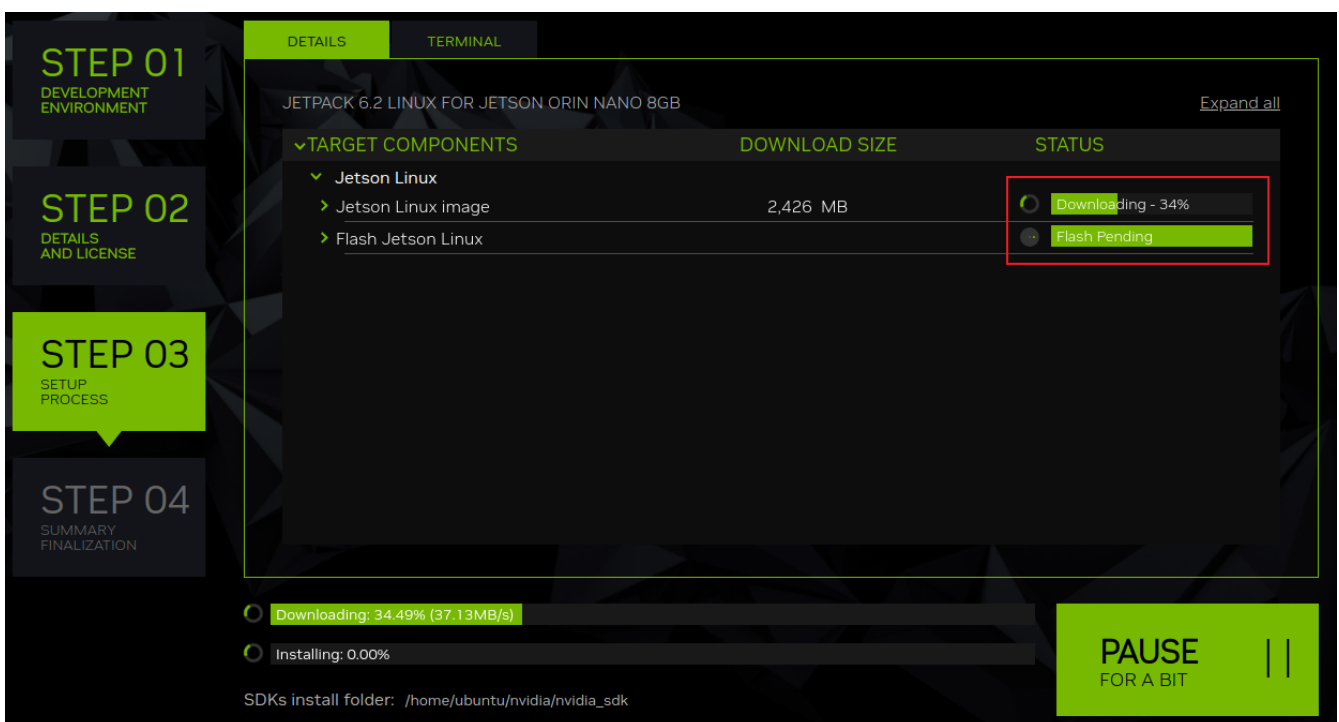
Note

The C1902 is fully compatible with the Developer Kit firmware and can be flashed directly with the official JetPack.

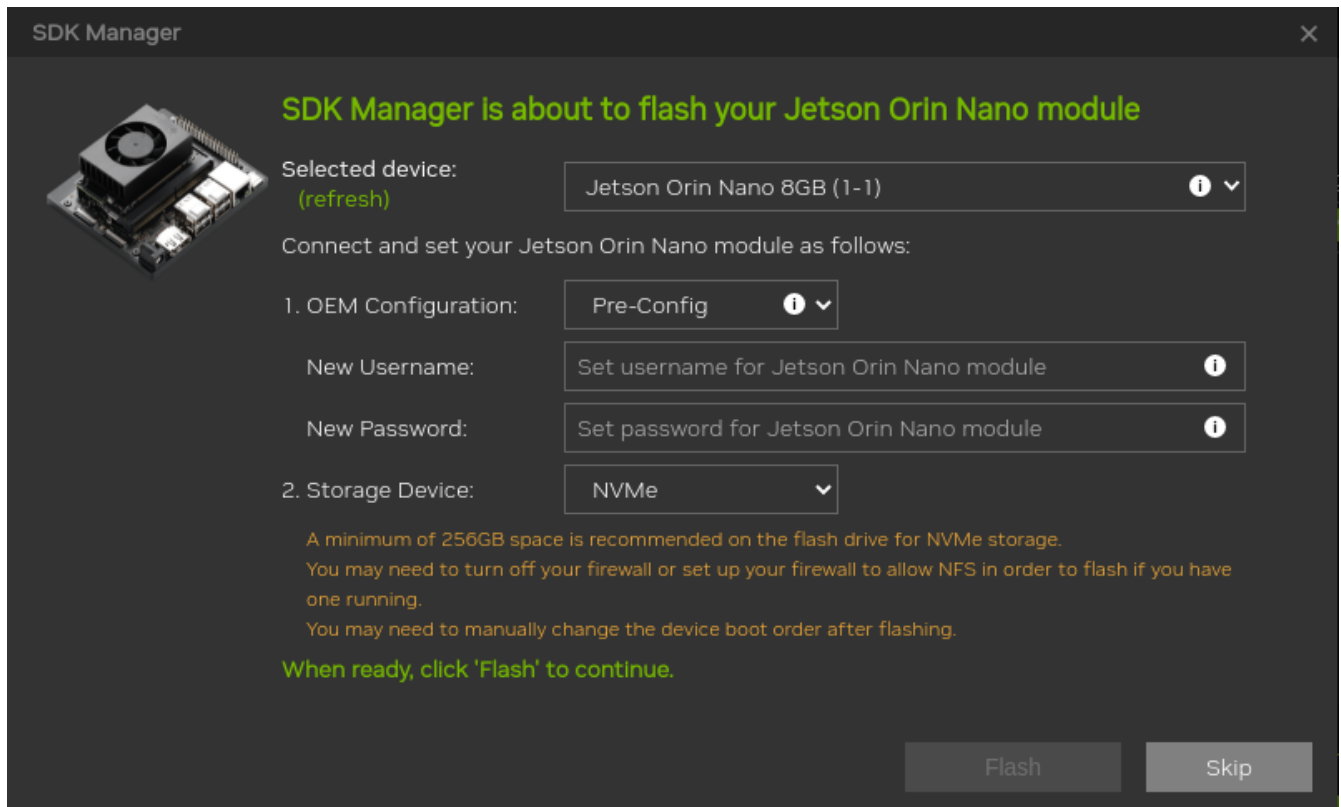
Uncheck Host Machine and keep only Target Hardware.



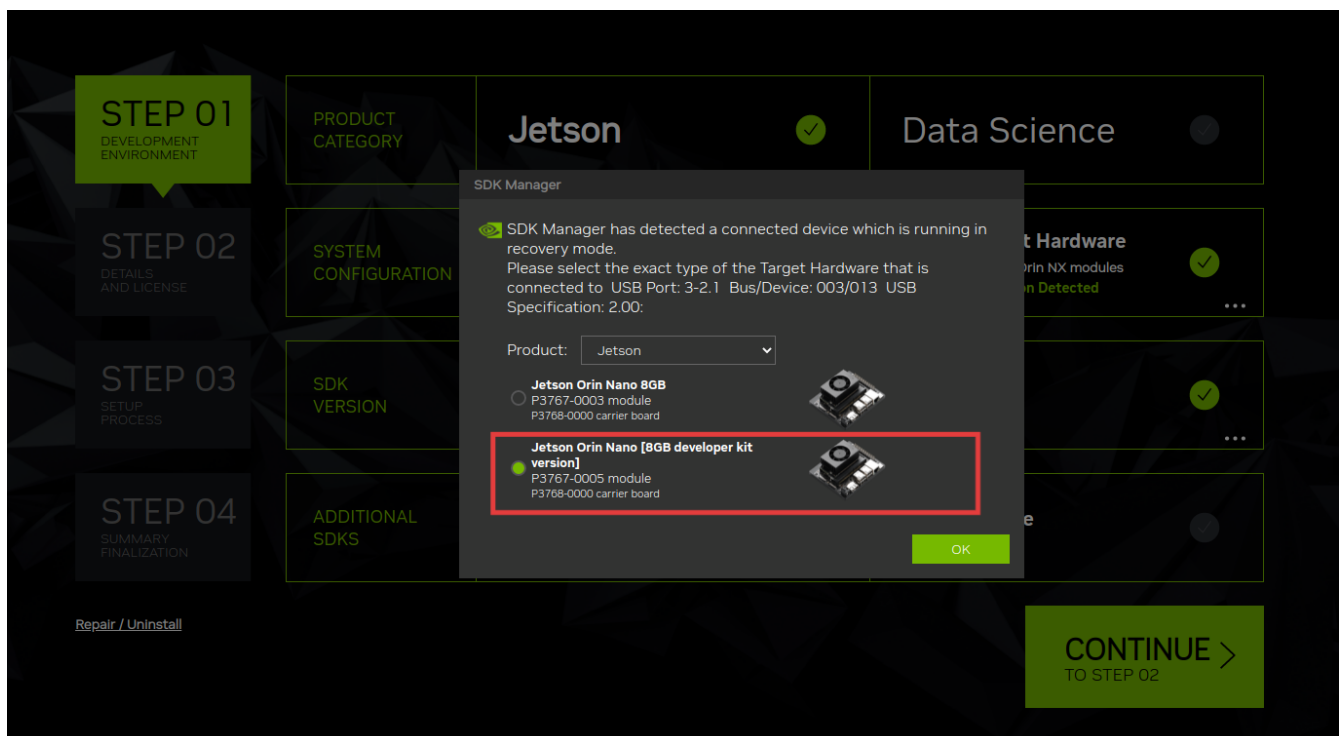
After selecting the JetPack version, start the flashing flow.



In the flashing configuration window, select the storage medium and system version.



Click Flash to start and wait for it to finish.



4. Finish

After flashing, the C1902 reboots automatically into the system. The first boot requires initial system configuration.



First Boot Configuration

- Select language, time zone, and keyboard layout
- Set a username and password
- Once done, you reach the Ubuntu desktop

Troubleshooting

Cannot Enter Recovery Mode

- Confirm the FC REC and GND pins are shorted correctly
- Confirm the power supply is working
- Check that the USB Type-C cable is a data cable

SDK Manager Cannot Detect the Device

- In VMware, confirm the USB device is connected to the VM
- Run `lsusb` to confirm the device is detected
- Try a different USB port

Flashing Failed

- Ensure enough disk space
- Keep the USB connection stable during flashing
- Re-enter recovery mode and retry

Next Steps

- [C1902 Introduction](#)

Source: [Kytech \(Guangzhou\) Co., Ltd](#) — LinkZee Labs

Flash the C2401

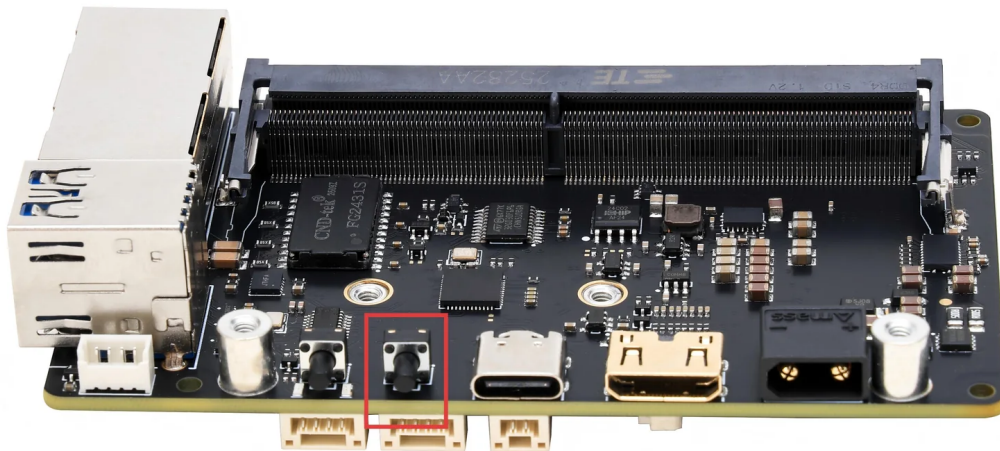
This guide explains how to flash the system onto the **C2401 Mini Kit**.

Before You Start

- Ubuntu VM and SDK Manager installed (see the [installation guide](#))
- A USB Type-C cable capable of data transfer
- C2401 Mini Kit

1. Enter Recovery Mode

The C2401 enters recovery mode through the **Recovery button** on its side.



Steps:

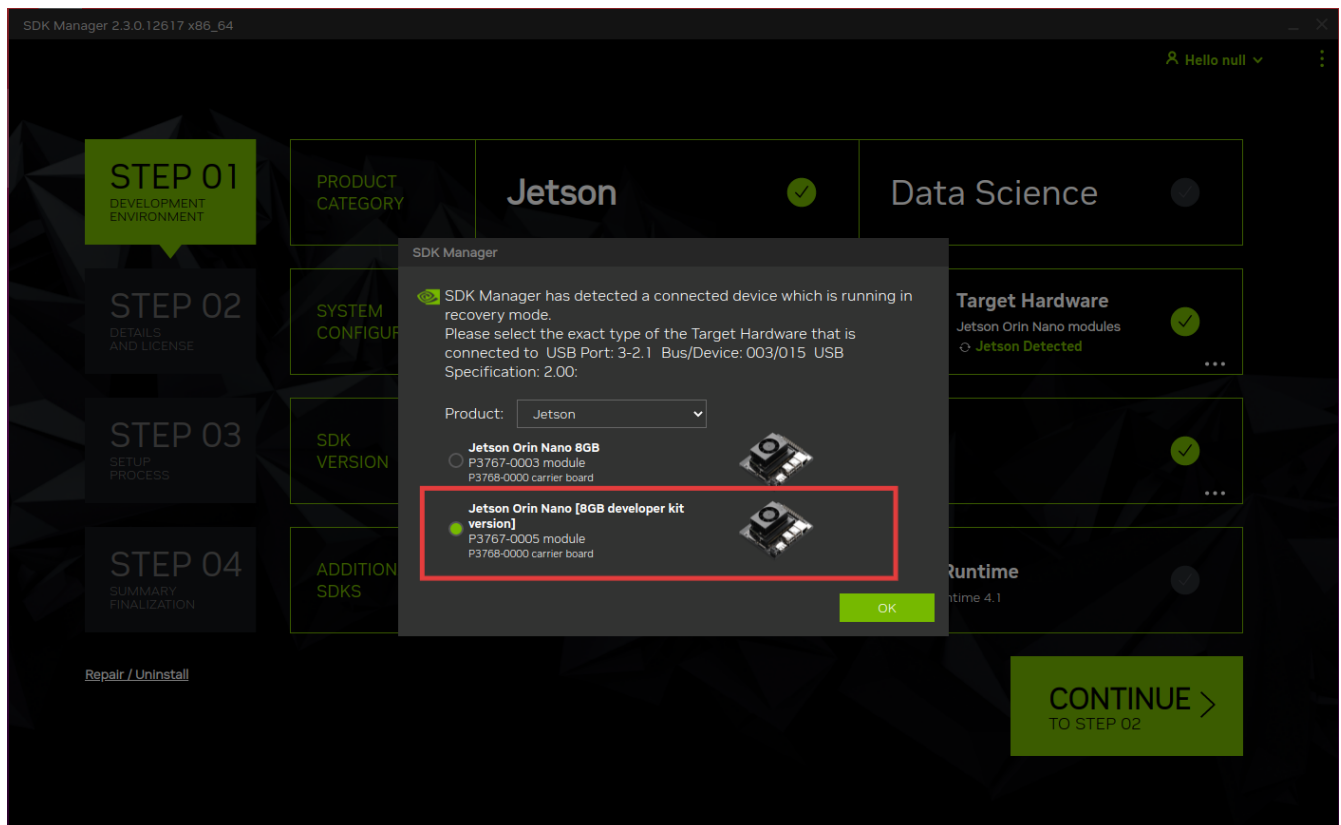
1. Disconnect the C2401 power.
2. Press and hold the **RECOVERY** button.
3. Connect the power.
4. Connect the C2401 to the VM with a USB Type-C cable.

5. Release the RECOVERY button.

2. Detect the Device

Run in the Ubuntu terminal:

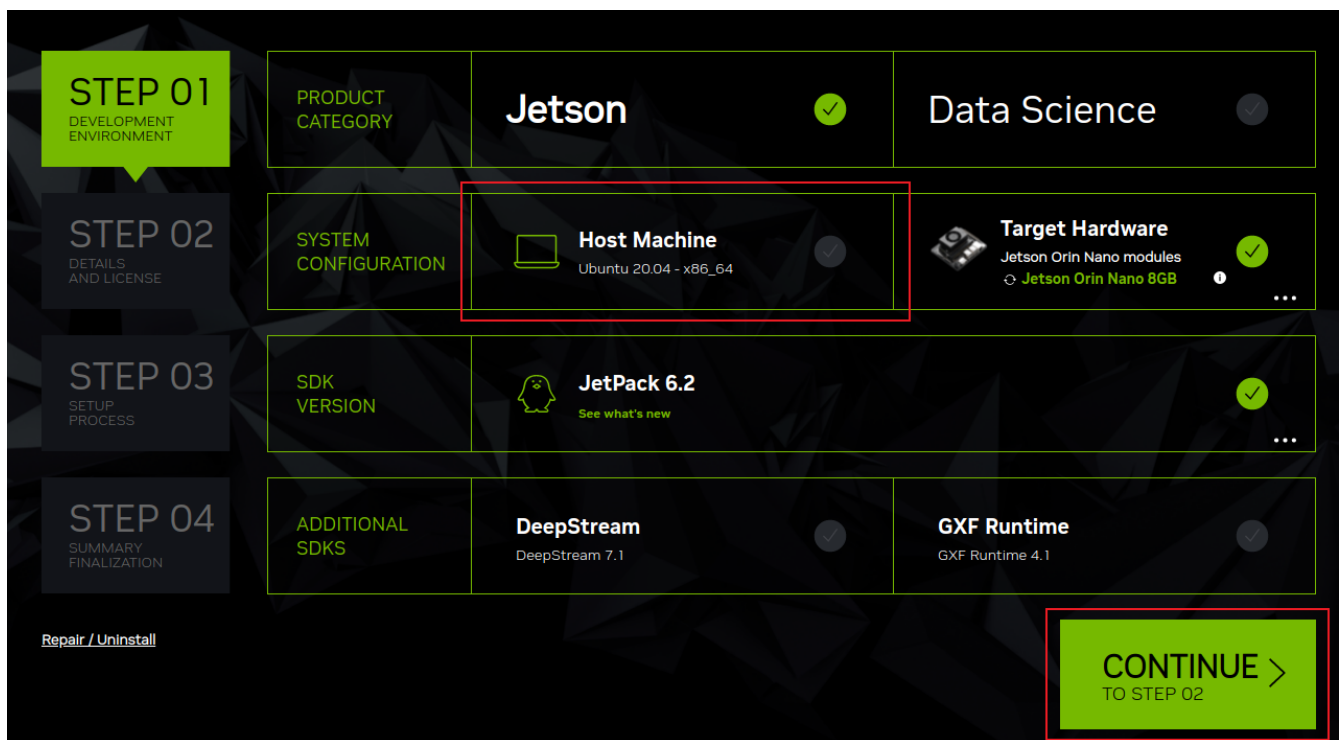
```
lsusb | grep -i nvidia
```



If an NVIDIA device is detected, recovery mode was entered successfully.

3. Flash with SDK Manager

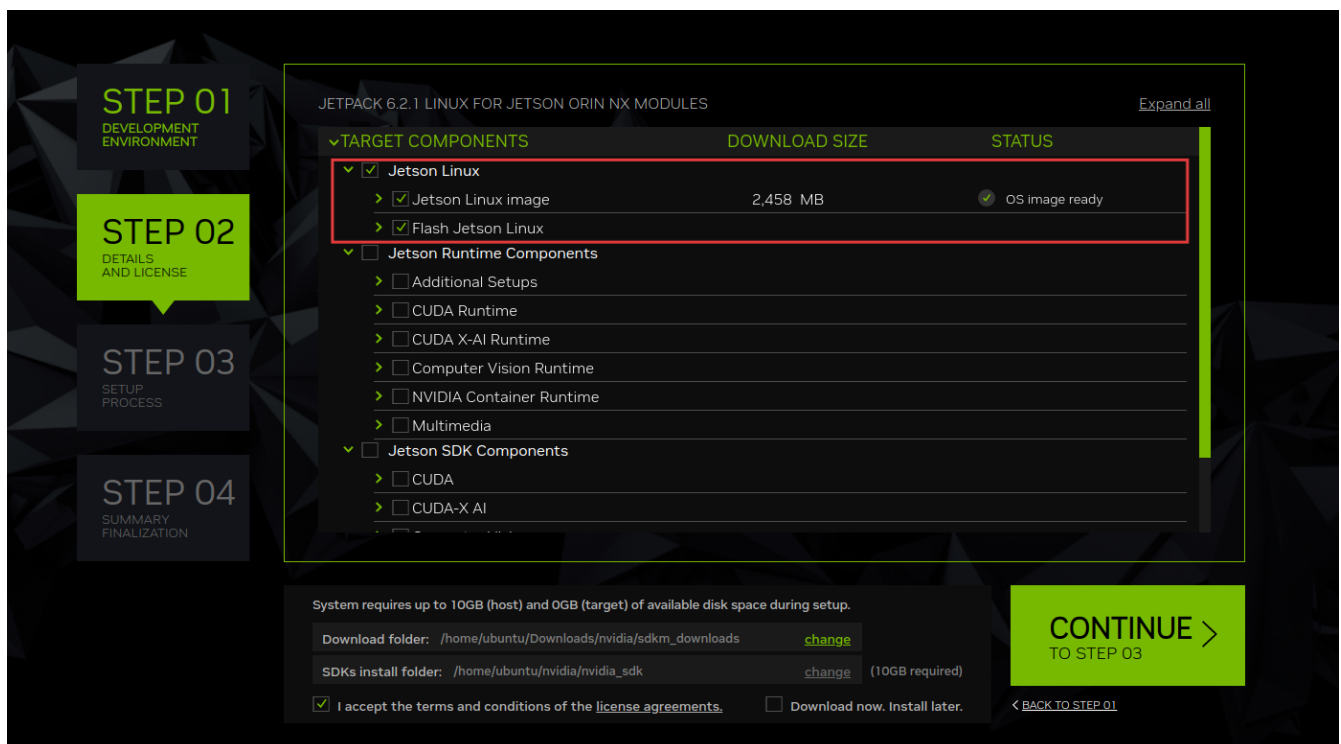
Open SDK Manager and select the corresponding module model.



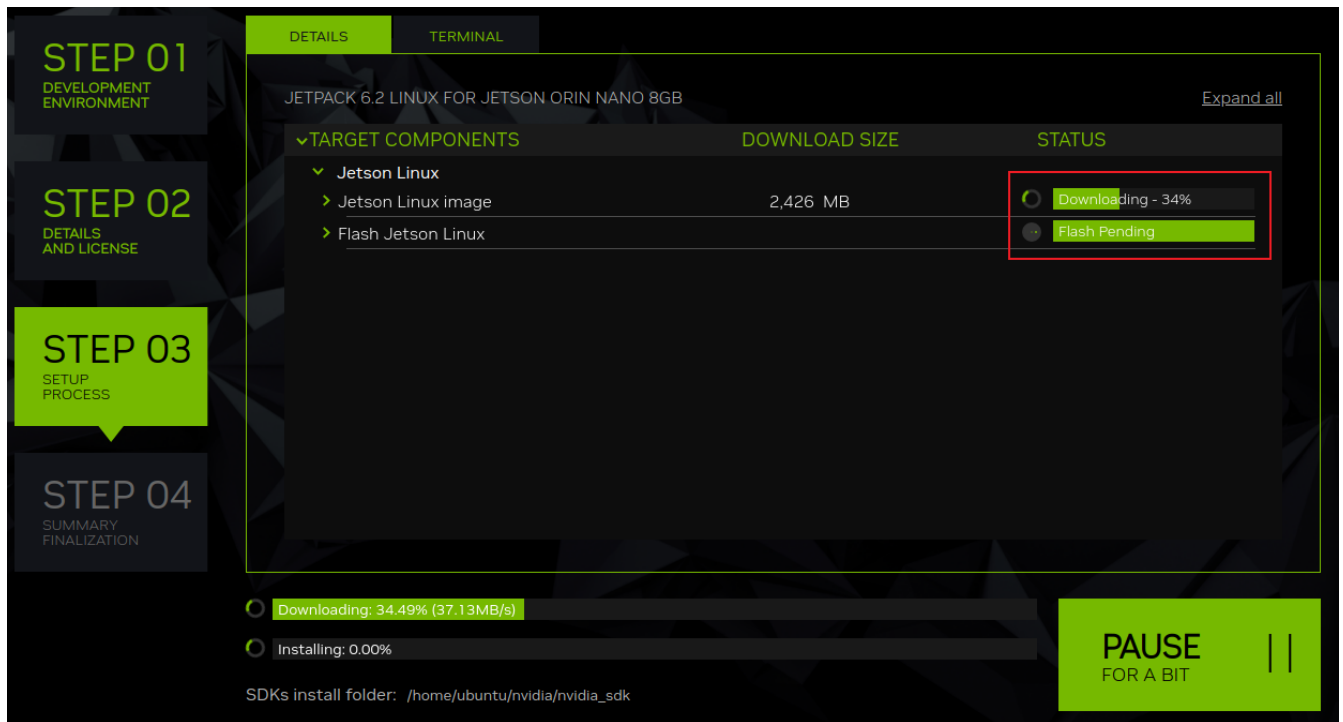
Note

The C2401 is an all-in-one Mini Kit; select the model that matches the built-in module.

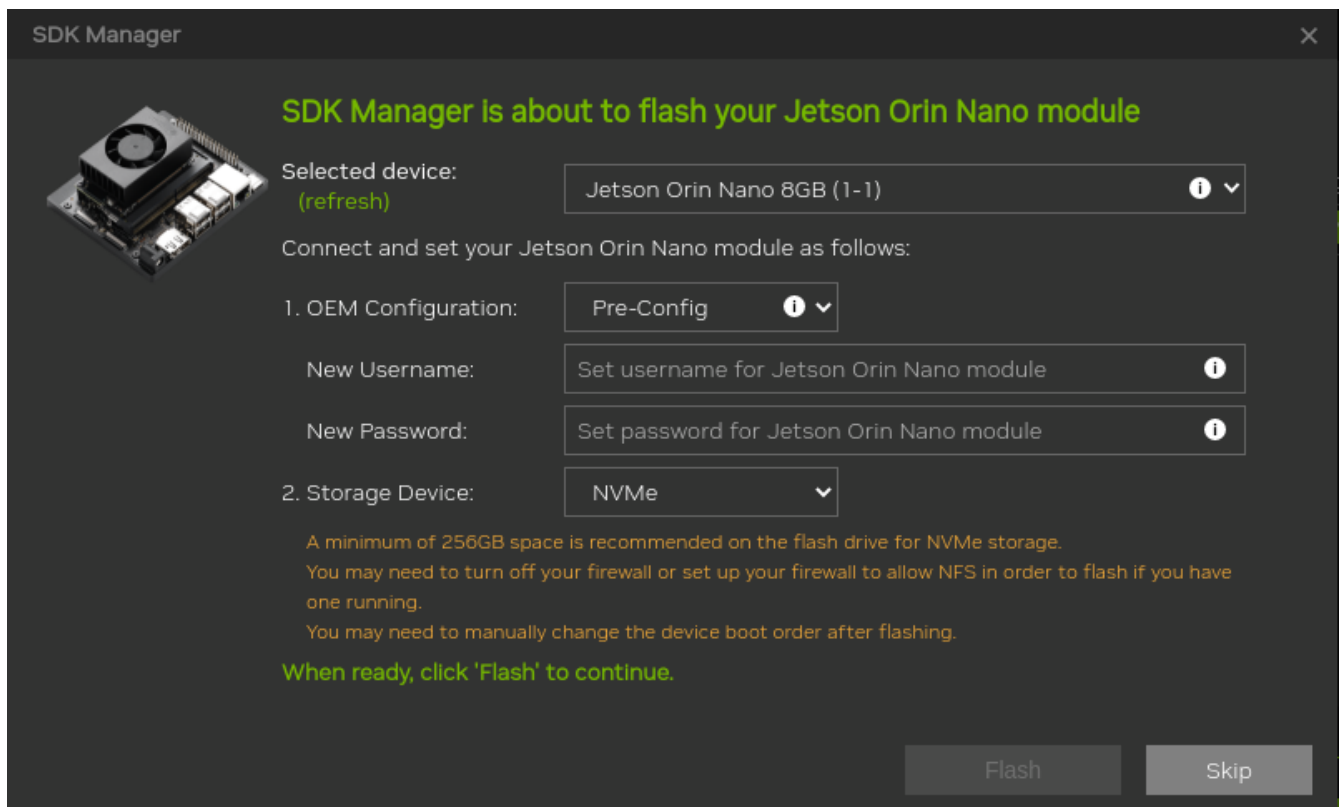
Uncheck Host Machine and keep only Target Hardware.



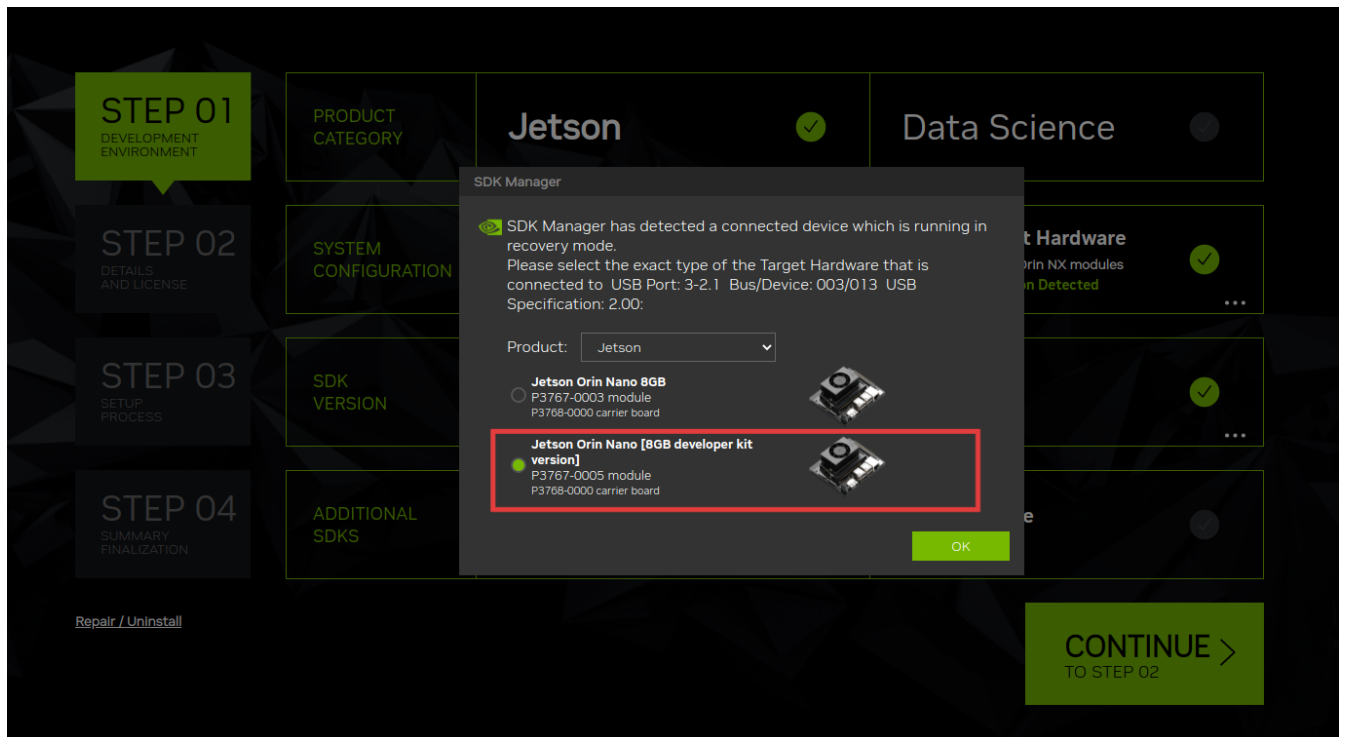
After selecting the JetPack version, start the flashing flow.



In the flashing configuration window, select the storage medium and system version.



Click Flash to start and wait for it to finish.



4. Finish

After flashing, the C2401 reboots automatically into the system.



First Boot Configuration

- Select language, time zone, and keyboard layout
- Set a username and password
- Once done, you reach the Ubuntu desktop

Troubleshooting

Cannot Enter Recovery Mode

- Confirm the RECOVERY button is pressed firmly
- Confirm the power supply is working
- Check that the USB Type-C cable is a data cable

SDK Manager Cannot Detect the Device

- In VMware, confirm the USB device is connected to the VM
- Run `lsusb` to confirm the device is detected
- Try a different USB port

Flashing Failed

- Ensure enough disk space
- Keep the USB connection stable during flashing
- Re-enter recovery mode and retry

Next Steps

- [C2401 Introduction](#)

Source: [Kytech \(Guangzhou\) Co., Ltd](#) — LinkZee Labs

Flash JetPack 7 (Orin Nano/NX)

Starting with **JetPack 7.2 (Jetson Linux r39.2, based on Ubuntu 24.04)**, the flashing method for Orin Nano/NX changed significantly: NVIDIA now uses a **Jetson ISO bootable-USB installer** that writes the OS to a microSD card or NVMe. **SD card images are no longer provided**, and SDK Manager is no longer required.

This guide covers the new ISO installer method, plus notes on third-party carrier boards (C1901/C1902) and SUPER mode.

Prerequisite: JetPack 6.x-generation firmware required

Installing JetPack 7.2 requires the device to already have **JetPack 6.x-generation UEFI/QSPI firmware (version 36.0 or newer)**.

- If your firmware is ≥ 36.0 , follow this guide directly.
- If it is **older than 36.0** (e.g. original factory firmware), you must **complete a JetPack 6.x install first** to bring the firmware to 36.x, then flash JetPack 7. See [Flash the Official Developer Kit](#) or the matching carrier-board guide to install JetPack 6.x first.

Scope

- **Modules:** Jetson Orin Nano / Orin NX
- **Version:** JetPack 7.2 (L4T r39.2)
- **Boards:** Official Developer Kit, plus reference-design third-party carrier boards (e.g. C1901, C1902). Third-party boards also need JetPack 6.x firmware first.

Overall Flow

The process has a **PC-side** part (download and create the USB) and a **device-side** part (boot, install, and set up). Confirm the firmware path before you start:

flowchart TD

Start([Start]) --> S1[Step 1
Confirm Jetson UEFI/QSPI firmware]

S1 -->|Firmware older < 36.0| UP[JetPack 6.x Update Path
Update UEFI/QSPI firmware]

UP --> Ready

S1 -->|Firmware ready ≥ 36.0| Ready{{Firmware ready}}

subgraph PC [PC side]

S2[Step 2 • JetPack 7.2
Download Jetson ISO]

S3[Step 3 • JetPack 7.2
Create Jetson ISO USB]

S2 --> S3

end

subgraph DEV [Device side]

S4[Step 4
Boot ISO, install Jetson Linux]

S5[Step 5
Boot Jetson Linux, finish setup]

S4 --> S5

end

Ready --> S2

S3 --> S4

S5 --> Done([Done])

Maps to the sections below: Step 1 → Section 2, Steps 2–3 → Sections 1 & 3, Step 4 → Section 4, Step 5 → Section 5.

1. Preparation

Item	Notes
Jetson ISO image	<code>jetsoninstaller-r39.2.0-*-arm64.iso</code> (r39.2)
USB flash drive	Used to create the ISO installer (install media, not the system drive)
Target storage	microSD (64GB UHS-1 or larger) or NVMe SSD (recommended, more capacity and speed)
Balena Etcher	Tool to write the ISO to the USB drive

Download the Jetson ISO (r39.2):

- Official download page: <https://developer.nvidia.com/embedded/jetpack/downloads>
- Direct link example: https://developer.nvidia.com/downloads/embedded/L4T/r39_Release_v2.0/iso/jetsoninstaller-r39.2.0-*-arm64.iso

Choose one target storage:



microSD (64GB UHS-1 or larger)

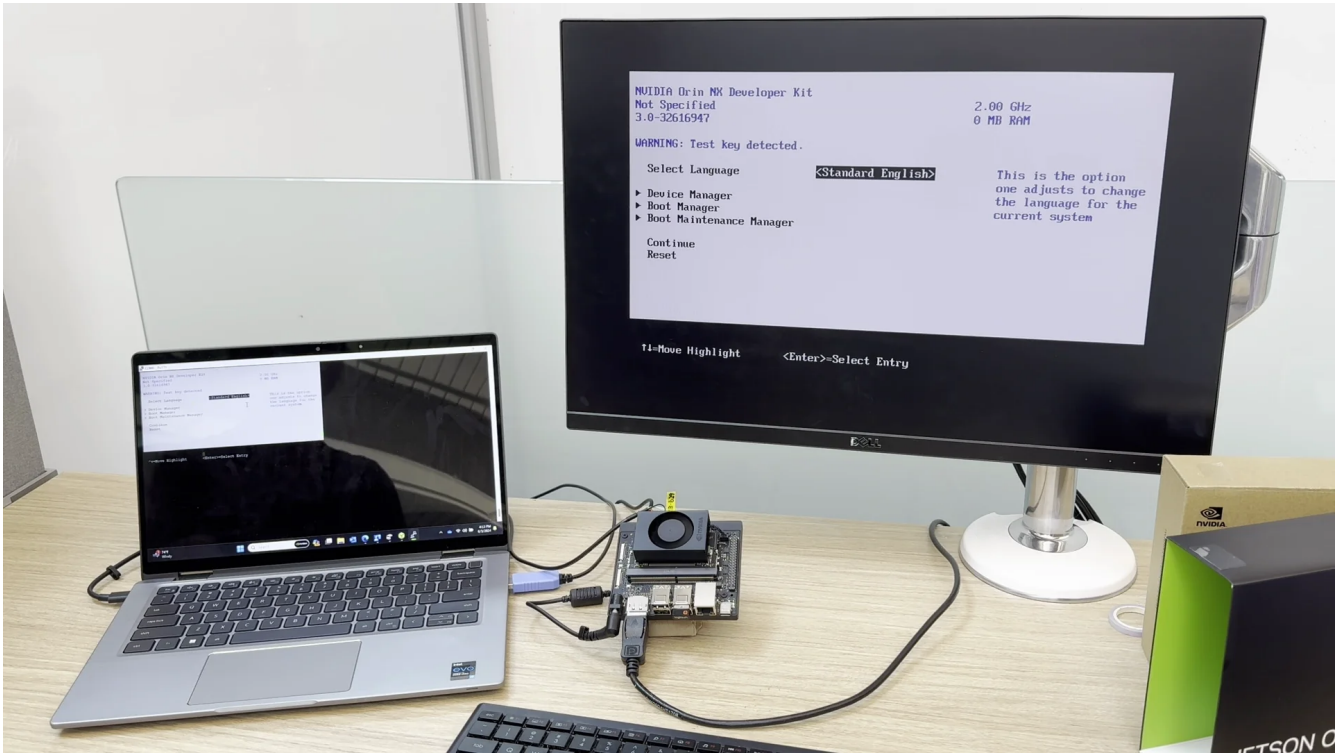


NVMe SSD (recommended)

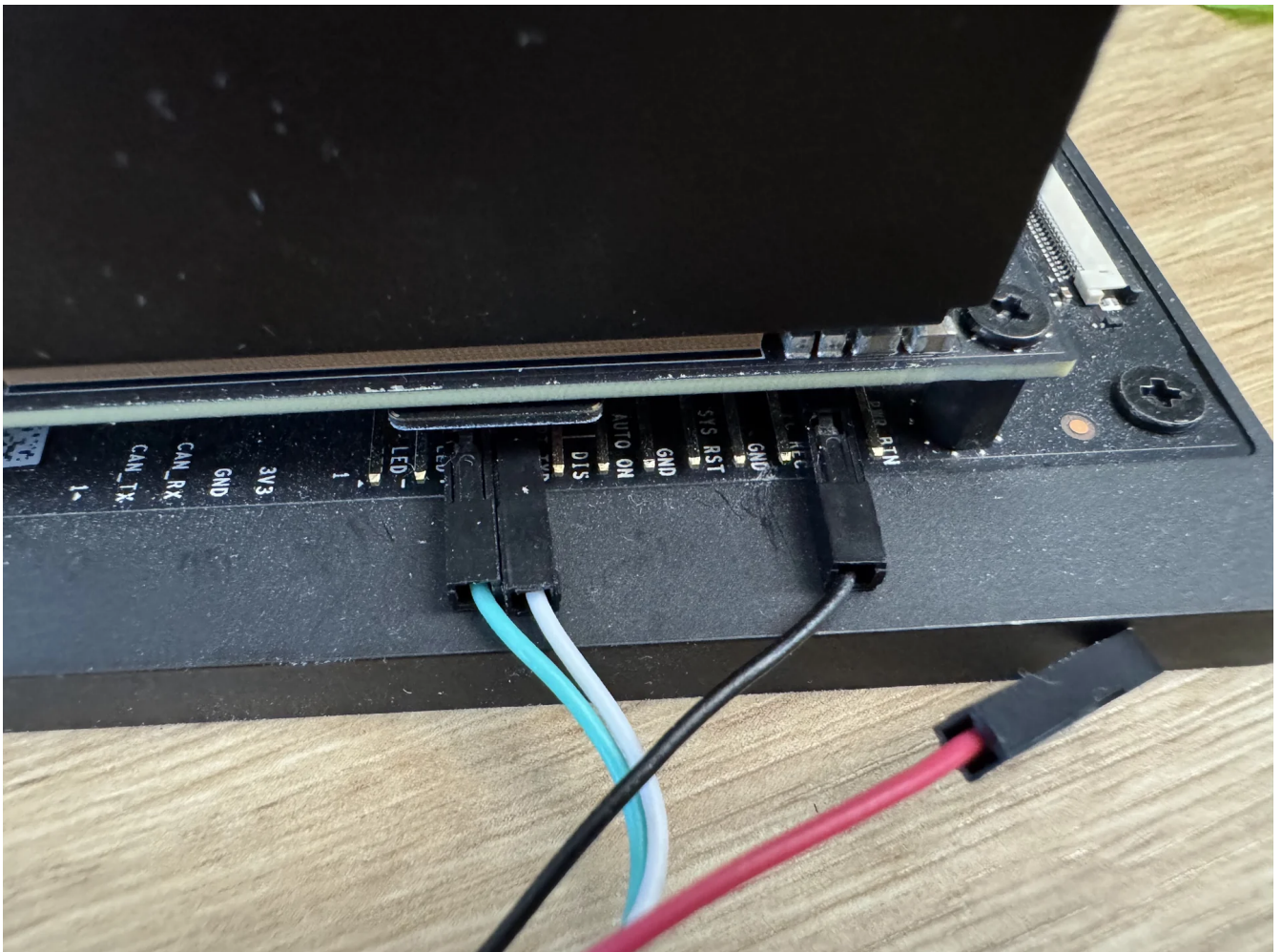
2. Confirm the Current Firmware Version

Before flashing JetPack 7, confirm the device's UEFI/QSPI firmware is ≥ 36.0 . Use any one of these methods:

1. **With a monitor:** Connect a DisplayPort monitor and USB keyboard, power on, repeatedly press **Esc** at the NVIDIA boot splash to enter the UEFI setup menu, and read the firmware version.
2. **Headless serial:** Connect a USB-to-TTL serial cable to the Button Header (RXD = pin 3, TXD = pin 4, GND = pin 7), open a serial console, power on, and press **Esc** repeatedly to enter the UEFI menu.
3. **Boot media test:** Simply try booting the JetPack 7 install media (less precise; not recommended for an exact check).



Reading the firmware version in the UEFI setup menu



Headless: a USB-to-TTL serial cable on the Button Header (RXD/TXD/GND)

If the version is older than **36.0** , complete the JetPack 6.x update from the "Prerequisite" box above before continuing.

3. Create the ISO USB

1. Open **Balena Etcher**.
2. Select the downloaded `jetsoninstaller-r39.2.0-*-arm64.iso`.
3. Select the USB drive to write.
4. Click **Flash** to write. When done you have a Jetson ISO installer USB.

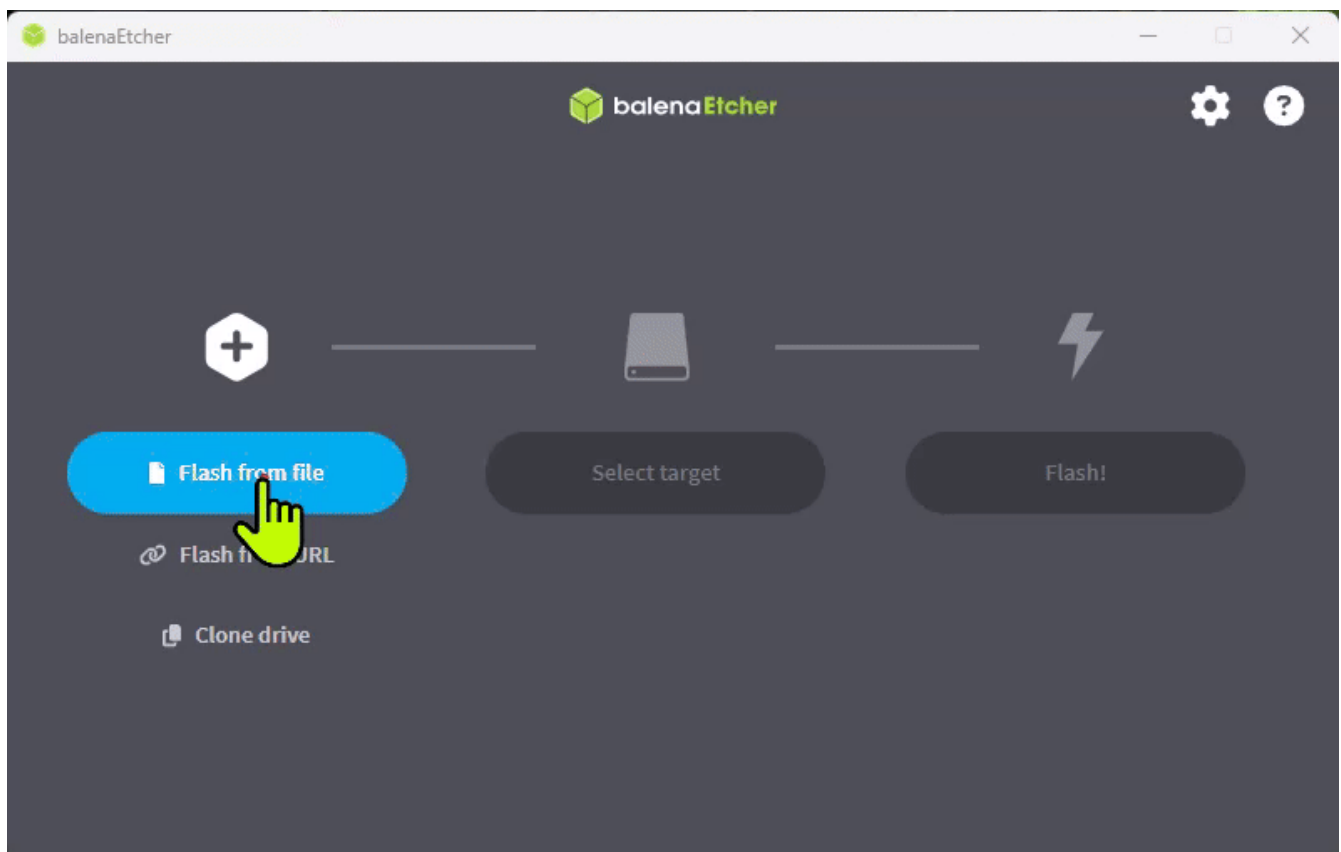
DOWNLOAD

Download Etcher

ASSET	OS	ARCH	
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Balena Etcher



Select the ISO and the USB drive, then click Flash

4. Boot the ISO and Install

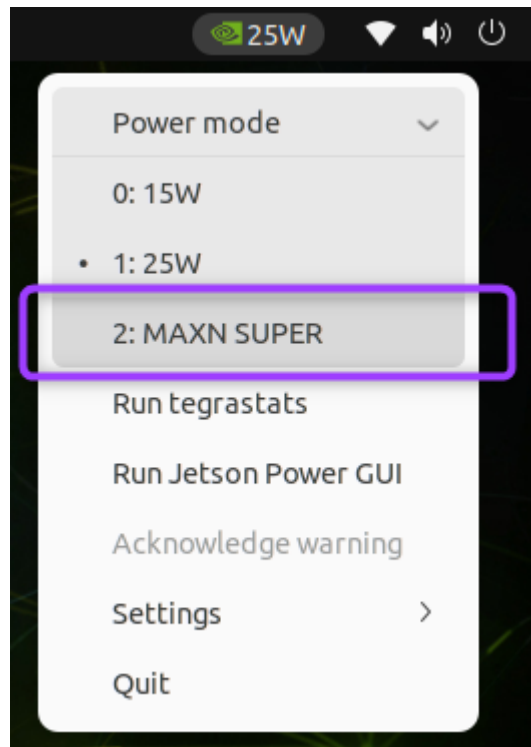
1. Power off, insert the **ISO installer USB**, install the target storage (microSD or NVMe), and connect a monitor, keyboard, and power.
2. Power on and follow the installer: - When the **UEFI capsule (firmware) update** confirmation appears, press **Y** (about a 30-second timeout). - The device runs a **dual-pass UEFI capsule update** and reboots automatically in between — this is normal; do not cut power. - At the **GRUB menu**, select **Install Jetson ISO r39.2**. - Choose the target storage device (**microSD** or **NVMe**). - Confirm the install (this **erases the target storage**).
3. When done, **remove the USB**, reboot, and boot from the target storage.
4. Follow the prompts to complete the Ubuntu initial setup (language, time zone, keyboard, username, password).

5. Enable SUPER Mode

On first boot the default power mode is usually **25W**. Enabling maximum performance (SUPER mode) is **exactly the same as JetPack 6.2**:

1. Click the current power mode in the Ubuntu desktop top bar.
2. Select **Power Mode**.

3. Choose **MAXN SUPER**.



Selecting MAXN SUPER in the power mode menu

Seeing the **25W** and **MAXN SUPER** options means SUPER mode is active (normal mode only has 7W / 15W).

Other Flashing Methods (Optional)

The ISO installer is the method NVIDIA recommends, but the following two still work on JetPack 7.2 and suit headless, batch, or third-party-carrier-board scenarios:

SDK Manager (Direct Flash)

On an Ubuntu host, use NVIDIA SDK Manager's **Direct Flash** to install the OS and the JetPack components (CUDA, cuDNN, TensorRT, etc.) in one guided flow. For host and SDK Manager setup, see [Install Ubuntu VM and SDK Manager](#).

Command-line initrd flash (third-party boards / SUPER firmware)

When Orin Nano/NX uses NVMe as external storage, NVIDIA recommends the `l4t_initrd_flash.sh` command-line flash. **Flashing SUPER firmware on a third-party board uses the exact same command as JetPack 6.2 — only change the JetPack directory version to 7.2** (this relies on the official firmware cache, so complete at least one full flash with SDK Manager first).

Put the board into recovery mode (on C1901/C1902, short **FC REC** and **GND** with a jumper, connect Type-C to the host, and power on), close any running SDK Manager, then run:

```
cd /home/ubuntu/nvidia/nvidia_sdk/JetPack_7.2_Linux_JETSON_ORIN_NANO_TARGETS/Linux_for_Tegra
sudo ./tools/kernel_flash/l4t_initrd_flash.sh --external-device nvme0n1p1 \
-c tools/kernel_flash/flash_l4t_t234_nvme.xml -p "-c bootloader/generic/cfg/flash_t234_qspi.xml" \
--showlogs --network usb0 jetson-orin-nano-devkit-super internal
```

The only difference from JetPack 6.2 is that `JetPack_6.2.1_...` in the path becomes `JetPack_7.2_...`; every other parameter (including `jetson-orin-nano-devkit-super`) stays the same. For full background and recovery-mode details, see the "Flash SUPER firmware via command line" section in [Flash the C1902](#).

Troubleshooting

- **Stuck on firmware update / repeated reboots:** The dual-pass capsule update reboots several times — this is normal; never cut power mid-way.
- **No Install option in GRUB / cannot boot the ISO:** Confirm the firmware is ≥ 36.0 (see Section 2); older firmware needs the JetPack 6.x update first.
- **Target storage not found:** Confirm the microSD/NVMe is properly installed and detected; a single-sided NVMe SSD is recommended.

Source: [NVIDIA Jetson Orin Nano Developer Kit — Quick Start Guide](#)